Game-theoretic Bandits for Network Optimization with High-probability Swap-regret Upper Bounds

Zhiming Huang and Jianping Pan

Abstract

In this paper, we study a multi-agent bandit problem in an unknown general-sum game repeated for a number of rounds (i.e., learning in a black-box game with bandit feedback), where a set of agents have no information about the underlying game structure and cannot observe each other's actions and rewards. In each round, each agent needs to play an arm (i.e., action) from a (possibly different) arm set (i.e., action set), and *only* receives the reward of the *played* arm that is affected by other agents' actions. The objective of each agent is to minimize her own cumulative swap regret, where the swap regret is a generic performance measure for online learning algorithms. Many network optimization problems can be cast with the framework of this multi-agent bandit problem, such as wireless medium access control and end-to-end congestion control. We propose an online-mirror-descent-based algorithm and provide high-probability swap-regret upper bounds based on refined martingale analyses, which can further bound the expected swap regret instead of the pseudo-regret studied in the literature. Moreover, the high-probability bounds guarantee that correlated equilibria can be achieved in a polynomial number of rounds if the algorithms are played by all agents. To assess the performance of the studied algorithm, we conducted numerical experiments in the context of wireless medium access control, and we performed emulation experiments by implementing the studied algorithms through the Linux Kernel for the end-to-end congestion control.

Index Terms

Multi-agent bandits, game theory, equilibrium learning, correlated equilibrium, swap regret

I. Introduction

The *multi-armed bandit (MAB)* is a theoretical model for online learning problems. The name comes from imagining a gambler needs to play one of the arms on a slot machine in each round. If an arm is played, then the gambler will receive a random reward. The objective of the gambler is to accumulate as many rewards as possible within T rounds. As the information about which arm can return the highest rewards is not prior knowledge, the gambler faces a dilemma in each round between playing the currently best arm (i.e., exploitation) or playing other arms to learn more about their rewards (i.e., exploration).

To adapt to more complex scenarios in reality, many variants of MABs have been proposed. In this paper, we study a variant called *multi-agent bandits in an unknown general-sum game (MAB-UG)*. This variant draws its motivation from many practical network optimization problems, such as wireless access control and end-to-end congestion control in computer networks. In the problem of wireless medium access control, a set of devices need to access a shared communication channel to send packets in each time slot without collisions. In the case of end-to-end congestion control, each host has no information about others and needs to choose a transmission rate or congestion window, hoping to maximize its throughput without congesting the network. MAB-UG can effectively encapsulate the dynamics due to competition and information constraints in these network optimization problems, offering an opportunity to develop innovative algorithms from a game-theoretic learning perspective.

The MAB-UG setting can be referred to as the black-box game studied in [3], where a set of agents $\mathcal{N}$, each associated with K_n (possibly different) arms (i.e., actions), are playing an unknown general-sum game repeated for T rounds. All agents have no information about the structure of the underlying game and cannot observe each other's actions and rewards. In each round, each agent needs to play an arm a_n^t from an arm set A_n, and observes the corresponding reward/loss.

The only information is the observed reward/loss for their own played arm in each round. Thus, each agent faces a non-stochastic multi-armed bandit problem with adaptive (i.e., non-oblivious) adversaries. The objective for each agent is to accumulate as many rewards as possible and the empirical joint distribution of all agents' actions reaches an ϵ-*correlated equilibrium (CE)* [4], a concept more general than the well-known Nash equilibrium, within T rounds. Intuitively, the ϵ-CE is a state that the expected incentives (e.g., the reward difference) for each agent to deviate from a suggested action are no more than $\epsilon \geq 0$, where the expectation is taken with respect to the joint distribution of all agents' actions.

As each agent has limited knowledge about the environment and can only learn from the rewards of the played arm affected by others' actions in each round, the algorithm to address MAB-UG must be carefully designed to balance the tradeoff between exploration and exploitation. The performance of an algorithm is usually measured by *regret*. The most oft-used definition of regret in the bandit literature is called the *external regret* [5], [6], which measures the performance loss of an algorithm against

A preliminary version of this paper has been published in the 39th conference on Uncertainty in Artificial Intelligence (UAI 2023) [1], and the part of emulation experiments of this paper has been published in the 43rd IEEE International Conference on Distributed Computing Systems (ICDCS 2023) [2]. In this paper, we study a generalized version of the algorithms studied in the preliminary version and provide high-probability upper bounds for swap regret.

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a set of competitors always playing a fixed action. However, minimizing the external regret is not enough for MAB-UG, as another objective is to achieve the ϵ-CE. Fortunately, it is proved in [7], [5] that if every agent plays an algorithm that minimizes *internal regret*, then the empirical joint distribution of actions converges to an ϵ-CE. The internal regret is the performance loss for an algorithm that plays arm a instead of playing another arm a'. In this paper, we study a stronger regret notion called *swap regret* introduced by [8], which is a generalization of the above two regrets, comparing the performance of a learning algorithm against a larger set of competitors. The swap regret uses swap functions F that take the arms played by an algorithm as input and output the arms to be compared. Thus, by changing the swap functions, the swap regret can boil down to external and internal regret.

The swap regret has been extensively studied in terms of *pseudo-regret* (or *weak regret*) [8], [9], [10], i.e.,

$$\max_F \mathbf{E}\left[\sum_{t=1}^{T}\sum_{a\in A_n}\mathbf{1}[a_n^t = a]r_{a,F(a)}\right],$$

where $\mathbf{1}[a_n^t = a]$ is the indicator function that returns 1 if arm a is the played arm a_n^t in round t , and $r_{a,F(a)}$ is the instantaneous swap regret with respect to arm a and swap function F (see Sec. III for detailed definitions). The *conditionally expected swap regret* is studied in [11], i.e.,

$$\max_F \sum_{t=1}^{T}\sum_{a\in A_n} p_a^t r_{a,F(a)},$$

where p_a^t is the expectation of $\mathbf{1}[a_n^t = a]$ conditioned on the history in the past $t-1$ rounds. The conditionally expected swap regret also bounds the pseudo-regret by taking expectations on the randomness of algorithms. However, bounding the above regret can only guarantee the expected swap regret

$$\mathbf{E}\left[\max_F \sum_{t=1}^{T}\sum_{a\in A_n}\mathbf{1}[a_n^t = a]r_{a,F(a)}\right]$$

is bounded when adversaries are not adaptive (i.e., oblivious) [12], but each agent in MAB-UG is facing other agents as adaptive adversaries. Thus, a more meaningful but challenging bound is on the instantaneous swap regret

$$\max_F \sum_{t=1}^{T}\sum_{a\in A_n}\mathbf{1}[a_n^t = a]r_{a,F(a)}$$

for any sequence of actions and rewards, which is helpful not only to equilibrium convergence but also to the bound of the expected swap regret with respect to all agents' randomness.

The main contributions of our work are as follows. First, we give an instantaneous swap regret analysis for an *online-mirror-descent (OMD)*-based algorithm to address MAB-UG, called *OMD learning for CE with implicit exploration (OMD-LCE-IX)*. OMD-LCE-IX is based on the swap-regret-minimizing framework proposed by [8], and the main idea is to call K_n OMD algorithms with the Implicit eXploration (IX) technique [13], [14] as subroutines. Then, the probability of selecting an arm is obtained by the Markov steady-state distribution of the Markov process among K_n subroutines, and the reward/loss is proportionally fed to the subroutines for updates.

However, the existing concentration inequality for the IX technique cannot be simply applied to the analysis of swap regret. The main difficulties are twofold. First, while the swap regret can only be represented as the aggregate of external regrets of subroutine algorithms *in expectation*, this representation does not hold for the instantaneous swap regret. Second, the reward/loss of each arm results from all agents' actions, which is not determined at the beginning of each round as in the single-agent bandit setting (see more discussions in Sec. III).

To address this problem, we prove a novel general-form concentration inequality between the IX loss estimator and the swapped loss based on a refined martingale analysis by treating the K_n subroutine algorithms as a whole. Based on this concentration inequality, we show that with probability at least $1-\delta$ for $\delta\in(0,1)$, the leading term of the instantaneous swap regret of OMD-LCE-IX is bounded by $O(K_n\sqrt{T\ln(K_n/\delta)})$ for each agent $n\in\mathcal{N}$.

Furthermore, by integrating the tails of this high-probability bound for the instantaneous swap regret, we show that the leading term of the expected swap regret of OMD-LCE-IX is bounded by $O(K_n\sqrt{T\ln K_n})$ with respect to all agents' randomness. The leading term of the above swap-regret bounds is within a factor of $O(\sqrt{K_n})$ of the current swap-regret lower bound $\Omega(\sqrt{K_nT\ln K_n})$ by [10], although the tightness of the lower bound is still an open problem. It is also guaranteed that OMD-LCE-IX can converge to ϵ-correlated equilibria for unknown general-sum games in a polynomial number of rounds if the algorithm is played by all agents.

Moreover, we not only conduct numerical experiments to show that MAB-UG can be applied to the wireless medium access control, but also perform emulation experiments to showcase the end-to-end congestion control by implementing the studied algorithms through the Linux Kernel.

The rest of the paper is organized as follows. In Sec. II, we review the works that are most related to MAB-UG. The problem settings are described in Sec. III. The OMD-LCE-IX algorithm is proposed in Sec. IV, with analytical results presented in Sec. V. The numerical and emulation experiment results are shown, compared and discussed in Sec. VI and Sec. VII, respectively. Sec. VIII concludes the paper. The detailed proofs of the swap-regret upper bound are deferred to the Appendix.

II. Related Works

Multi-agent bandits: Multi-agent bandits consider a group of agents participating in decision making, and aim to improve learning efficiency through collaborations. The works about multi-agent bandits are mainly focused on improving rewards by communication [15], [16], [17], [18], identifying the best arm to avoid collision [19], [20], [21], [22], [23], and voting for playing arms [24]. All the above bandit settings assume the arm set for each agent is identical, and the reward for an agent does not depend on the actions of other agents, or just follows a simple collision model. MAB-UG considers (possibly) varied arm sets for different agents and more general competitions among agents.

Table I: Swap-regret bounds in the bandit settings

Leading upper bound, Computational cost, Regret notion	Lower bound
$O\left(\sqrt{TK_n^3}\right)$, poly-time, pseudo-regret [8]	$\Omega\left(\sqrt{TK_n}\right)$ [8]
$O\left(\sqrt{TK_n^2\ln(K_n)}\right)$, exp-time, pseudo-regret [9]	
$O\left(\sqrt{TK_n^2}\right)$, poly-time, pseudo-regret [10]	$\Omega\left(\sqrt{TK_n\ln K_n}\right)$ [10]
$O\left(\sqrt{TK_n^2\ln(K_n/\delta)}\right)$, poly-time, conditionally expected regret [11]	
$O\left(K_n\sqrt{T\ln(K_n/\delta)}\right)$, poly-time, instantaneous regret (our work, Theorem V.3)	
$O\left(K_n\sqrt{T\ln K_n}\right)$, poly-time, expected regret (our work, Corollary V.4)	

Learning in games: The history of learning in games can be traced back to the fictitious play for the two-player zero-sum games [25], [26]. Nevertheless, such a fictitious play requires that the decisions of opponents can be observed, and thus it cannot be applied to the unknown games where the agents can only observe their own outcomes (or rewards). To address the challenges of unknown games, online learning has been introduced by many works for specific games such as potential games [27], [28], [29], [30], and mean-field games [31], [32], [33]. However, the above solutions for specific games depend on corresponding properties (e.g., potential functions for potential games), and thus cannot be easily extended to the general-sum games. Thus, we focus on the learning in the unknown general-sum games (i.e., black-box games [3]), which provides theoretical foundations for learning in general-sum Markov games [34].

Regarding the unknown general-sum games, there are mainly two lines of research depending on the observability of rewards. If the reward of an action can be observed regardless of whether it is played or not, we call it the *full-information* model [5], and if only the reward of a played action can be observed, then it is the *partial-information* model (i.e., *bandit* feedback). Recent years have witnessed steady progress in learning general-sum games in the full-information model [35], [36], [37], [38], [39], [40]. However, the results for the full-information model cannot be easily extended to the partial-information model, as less information is observed in each round, which makes the partial information model more challenging. The first work that addressed the unknown general-sum games with bandit feedback is [41], where an exponential-weighting-based technique is proposed to minimize the external regret. However, it is typically one of the goals in general games to search for correlated equilibria, and it is shown in [5] that only minimizing external regret cannot achieve this goal.

The authors of [8] generalized the notion of external and internal regrets to the swap regret, and proposed a polynomial-time swap-regret-minimizing framework based on K_n external-regret-minimizing subalgorithms, where K_n is the number of arms. They proved that if the external pseudo-regret of each subalgorithm can be represented by a concave function $r(T)$, where T is the time horizon, and if the dependency on K_n is ignored in $r(T)$, the swap pseudo-regret of their proposed algorithm is $K_n \cdot r(T)$. For OMD-based subalgorithms, $r(T) = O(\sqrt{TK_n})$ [42], and thus the analysis of [8] gives a pseudo-regret bound of $O(K_n\sqrt{TK_n})$.

Later, this bound was improved by a new algorithm [9] to $O(K_n\sqrt{T\ln K_n})$ but with an exponential computation complexity. Furthermore, the author of [10] improved the upper bound for the swap pseudo-regret to $K_n \cdot r(T/K_n)$ with a polynomial-time algorithm by adding another layer of randomness to the original framework [8], where in each round only one subroutine is selected according to the calculated Markov steady distribution. The selected subroutine selects an arm, and the reward will be *entirely* fed to this subroutine algorithm for updates. The modified framework gives a pseudo-regret of $O(\sqrt{TK_n^2})$ for the OMD-based subalgorithms. It was also proved by [10] that the lower bound for swap regret is $\Omega\left(\sqrt{TK_n\ln K_n}\right)$, which is tight in the full-information models. However, it still remains an open problem whether the lower bound is tight in the bandit settings. The author of [11] proved a high-probability bound of $O(K_n\sqrt{T\ln(K_n/\delta)})$ for the conditionally expected swap regret, which can bound the pseudo-regret by integrating the tails.

More recently, in our preliminary work [1], we proved high-probability bounds of $O(K_n\sqrt{T\ln(K_n/\delta)})$ for the instantaneous swap regret, and proved the first bound of $O(K_n\sqrt{T\ln(K_n)})$ for the expected swap regret. Table I summarizes the literature on the swap-regret bounds in the bandit settings.

In this paper, we establish high-probability bounds whose leading term is $O(K_n\sqrt{T\ln(K_n/\delta)})$ for the instantaneous swap regret of the generalized OMD-LCE-IX algorithm. By integrating the corresponding tails, we also bound the leading term of the expected swap regret in $O(K_n\sqrt{T\ln K_n})$ with respect to all agents' randomness.

III. System Model and Problem Formulation

A. The MAB-UG Model

In MAB-UG, the reward of each agent's action will be affected by the actions of other agents, and each agent has no prior knowledge about the environment such as the number of agents, the reward of each action, and the actions of other agents. A simple example of MAB-UG with two agents and two arms for each agent is shown in Fig. 1, where in the current round, Agent 1 plays arm a and only observes a normalized reward of 0.8, and Agent 2 plays arm c and only observes a normalized reward of 0.2. Both agents have no information about the arm played by the other agent, nor the rewards of the arms that are not played.

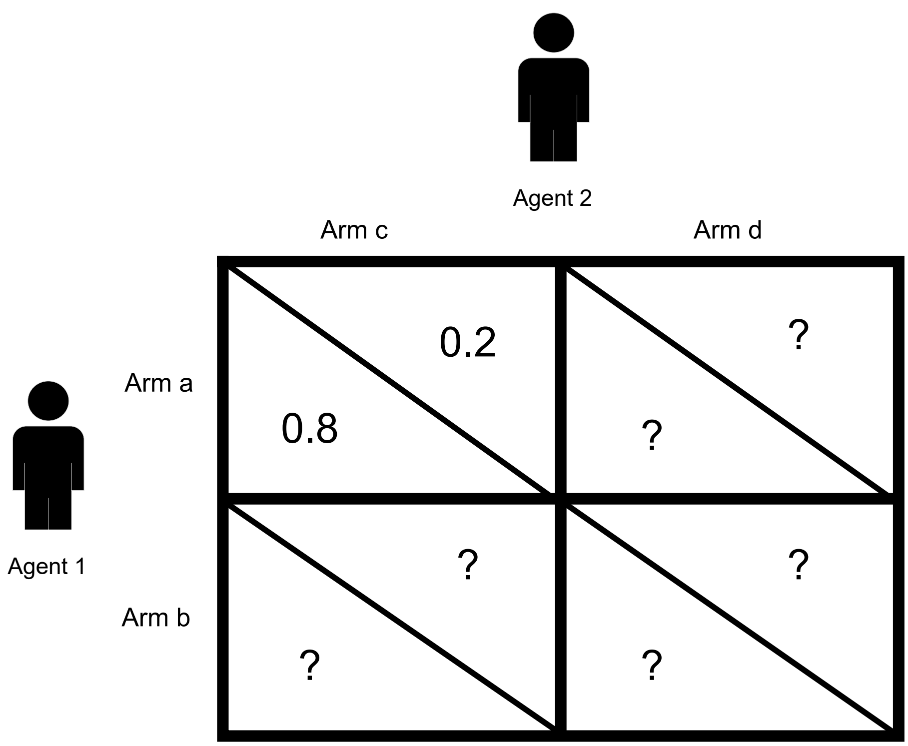


Figure 1: An example of MAB-UG with two agents and two arms for each agent.

Formally, let $\mathcal{N} := \{1, \ldots, N\}$ be the set of all agents and each agent $n \in \mathcal{N}$ is associated with a finite set of arms (i.e., actions) A_n with size K_n. The arm set for each agent is not required to be identical. Let $\mathcal{A} := \prod_{n\in\mathcal{N}} A_n$ be the space of all such arm sets, and $\mathbb{A} \in \mathcal{A}$ be an action profile (i.e., a vector of all agents' actions). The reward for agent n playing arm $a_n^t \in A_n$ in round t is determined by function $u_n : \mathcal{A} \to [0, 1]$, which maps the actions of all agents to agent n's rewards $u_n(a_n^t; \mathbb{A}_{-n}^t)$, where $(a_n^t; \mathbb{A}_{-n}^t)$ is an abbreviation of $\mathbb{A}^t := (a_1^t, \ldots, a_n^t, \ldots, a_N^t)$ with a highlight of agent n's action a_n against other agents' actions. Note that our algorithm and analyses also work for a time-varying reward function u_n^t. In addition, u_n^t can be determined in either an oblivious way or a non-oblivious (i.e., adaptive) way, corresponding to the oblivious adversary or the non-oblivious adversary in the single-agent bandits. In an oblivious way, $\{u_n^t\}_{t>0}$ is chosen at the beginning of the game, while in a non-oblivious way, each u_n^t is determined conditioned on all the agents' actions in the past.

One of the main differences between multi-agent bandits and single-agent bandits is the measurability of the rewards. If we are in the single-agent bandits, regardless of whether u_n^t is determined obliviously or non-obliviously, the reward of each arm in each round t is determined at the beginning of that round, before the agent plays an action. However, in the multi-agent bandits, as the reward of each arm for each agent is conditioned on other agents' actions, the reward of each arm in each round cannot be determined until all agents have played an action in that round.

Let $\mathcal{U} := \{u_1, \ldots, u_N\}$ be the set of reward functions for N agents. Note that neither $\mathcal{N}$ nor $\mathcal{U}$ is a prior knowledge to each agent, and each agent n only knows in advance her own set of arms A_n.

In each round $t = 1, \ldots, T$, each agent $n \in \mathcal{N}$ can use a *mixed* strategy to play an arm $a_n^t \in A_n$ according to a probability distribution over arms $P_n^t := \{p_a^t : \forall a \in A_n\}$, i.e., play arm $a \in A_n$ with probability p_a^t. Then, each agent n can only observe her own instantaneous reward $X_n^t := u_n(a_n^t; \mathbb{A}_{-n}^t)$.[1] Both the actions and the number of other agents cannot be observed. The objective of each agent is to accumulate as many rewards as possible over T rounds.

B. Problem Formulation

As each agent has little knowledge about the environment, it is inevitable for each agent to suffer a *regret*, i.e., the loss of rewards for not playing the optimal arm in hindsight that returns the highest cumulative rewards. In bandit problems, the problem of maximizing the cumulative reward is always converted to the problem of minimizing the regret. The notion of regret has many forms. The most oft-used regret in the bandit literature is the *external regret* [5]. Let $\mathbf{1}[a_n^t = a]$ be the indicator function that returns 1 if a is the played arm in round t and 0 otherwise. The external regret $R_n^{\text{ext}}(T)$ for agent n compares the cumulative reward of a learning algorithm with that of a set of competitors that always play a fixed arm up to round T, which is defined as follows:

$$R_n^{\text{ext}}(T) := \max_{a' \in A_n} \sum_{t=1}^{T} u_n(a'; \mathbb{A}_{-n}^t) - \sum_{t=1}^{T} \sum_{a \in A_n} \mathbf{1}[a_n^t = a] u_n(a; \mathbb{A}_{-n}^t),$$

However, only minimizing the external regret cannot guarantee the plays of agents will reach a *correlated equilibrium (CE)*. Therefore, we need a strictly stronger notion of regret that is the *internal regret*, which compares the actions of an agent in a pair-wise manner:

$$R_n^{\text{int}}(T) := \max_{a, a' \in A_n} \sum_{t=1}^{T} r_{(a,a'),n}^t, \tag{1}$$

where

$$r_{(a,a'),n}^t := \mathbf{1}[a_n^t = a] \left(u_n(a'; \mathbb{A}_{-n}^t) - u_n(a; \mathbb{A}_{-n}^t) \right)$$

is the instantaneous regret for agent n of having played arm a instead of arm a' in round t. As proved in [5], [7], by minimizing the internal regret for each agent, their empirical joint distributions of plays (i.e., $\hat{P}^T(\mathbb{A}) := \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}[\mathbb{A}^t = \mathbb{A}], \forall \mathbb{A} \in \mathcal{A}$) converge to an ϵ-CE, which is defined as follows.

Definition III.1. Let $\mathbf{P}$ be a joint probability distribution over $\mathcal{A}$. We say $\mathbf{P}$ is an ϵ-correlated equilibrium if the expected incentive for each agent n to deviate from action a to any other action $a' \in A_n$ is no more than $\epsilon \geq 0$, i.e., $\forall n \in \mathcal{N}$, we have

$$\sum_{(a; \mathbb{A}_{-n}) \in \mathcal{A}} \mathbf{P}((a; \mathbb{A}_{-n})) \left(u_n(a'; \mathbb{A}_{-n}) - u_n(a; \mathbb{A}_{-n}) \right) \leq \epsilon. \tag{2}$$

Note that $\mathbf{P}$ is the joint distribution, not the product distribution, which is the difference between the CE and the Nash equilibrium. When $\epsilon = 0$, $\mathbf{P}$ is the CE, which is more general than the well-known Nash equilibrium, as the CE does not require independence among actions. To give an intuition about the ϵ-CE, consider a case of congestion control in computer networks where a *mediator* (e.g., a router or switch) draws an action profile from $\mathbf{P}$ and privately recommends each action (e.g., the packet sending rate) to the corresponding host. If no host has an incentive of more than ϵ to choose a different action after seeing the recommended action, provided that other hosts follow the mediator's recommendation, then $\mathbf{P}$ is an ϵ-CE.

The concept of a *coarse correlated equilibrium (CCE)* also merits attention. This notion studies whether an agent is inclined to adhere to a predetermined action without changing it even after observing the mediator's recommendation. However, in the applications we are exploring—such as congestion control and wireless medium access control—agents have the flexibility to modify their decisions post-recommendation. Therefore, this paper will focus on the notion of CE rather than CCE.

Our objective is to achieve an ϵ-CE without a mediator by minimizing the internal regret for each agent. To achieve this objective, we consider a more general notion of regret, called *swap regret* [8], which can unify both the external regret and internal regret into the same framework by a swap function $F_n : A_n \to A_n$ that takes $a \in A_n$ as input and outputs $a' \in A_n$. Let $\mathcal{F}$ be a finite set of F_n. Then, the instantaneous swap regret for agent n with $\mathcal{F}$ up to round T is defined as follows:

$$R_n^{\text{swa}}(T, \mathcal{F}) = \max_{F \in \mathcal{F}} \sum_{t=1}^{T} \sum_{a \in A_n} \mathbf{1}[a_n^t = a] \left(u_n(F(a); \mathbb{A}_{-n}^t) - u_n(a; \mathbb{A}_{-n}^t) \right). \tag{3}$$

We can boil down the swap regret to the external regret by letting $\mathcal{F}$ be a set of K_n functions such that for any $a \in A_n$, $F_a \in \mathcal{F} : A_n \to a$. Similarly, the internal regret can be obtained by letting $\mathcal{F}$ be a set of $K_n(K_n - 1)$ functions such that for any pair of $a, a' \in A_n$, we have $F_{(a,a')}(a) = a'$ and $F_{(a,a')}(a'') = a''$ for any other $a'' \in A_n$. Thus, by minimizing the swap

[1] For the convenience of algorithm description and analysis, we sometimes use an equivalent notion called the instantaneous loss, i.e., $1 - X_n^t$, and denote by $y_{n,a}^t := 1 - u_n(a; \mathbb{A}_{-n}^t)$ the instantaneous loss function if agent n plays $a \in A_n$.

regret of a learning algorithm for a general $\mathcal{F}$ of any possible mappings F, we can: 1) show that the learning algorithm has a bounded performance gap from a broader range of competitors, 2) minimize the internal and external regrets simultaneously, and achieve the ϵ-CE for all agents.

IV. The OMD-LCE-IX Algorithm

The OMD-LCE-IX algorithm adopts the swap-regret-minimizing framework introduced by [8] and calls K_n OMD algorithms with the Tsallis entropy [42] and implicit exploration techniques [13], [14] as subroutines. Each subroutine maintains a meta-distribution, and the action selection probability is calculated from the meta-distributions. The observed reward or loss will be assigned proportionally to each subroutine to update the meta-distributions.

For each agent n, we define a meta-distribution $Q_a^t := \{q_{a,a'}^t : \forall a' \in A_n\}$ for each arm $a \in A_n$ such that $q_{a,a'}^t \in [0,1]$ and $\sum_{a' \in A_n} q_{a,a'}^t = 1$.

Each meta-distribution is managed by its corresponding OMD subroutine. Therefore, we use the same index, denoted by a, to identify each subroutine. Denote by $\mathbb{Q}_n^t := [Q_a^t]_{a \in A_n}$ the $K_n \times K_n$ matrix with each row being Q_a^t. Then, we determine the sample distribution P_n^t by solving the following equations:

$$P_n^t = P_n^t \mathbb{Q}_n^t, \tag{4}$$

where P_n^t is a row vector of $p_a^t, \forall a \in A_n$ and $\sum_{a \in A_n} p_a^t = 1$. That is, for each $a' \in A_n$, we have $p_{a'}^t = \sum_{a \in A_n} p_a^t q_{a,a'}^t$, which is similar to the calculation of the stationary distribution of a Markov process with the transition matrix being $\mathbb{Q}_n^t$. The intuition behind (4) is to make the probability of playing arm $a' \in A_n$ directly according to P_n^t equivalent to the probability of first sampling any arm $a \sim P_n^t$ and then playing a' according to Q_a^t.

The suffix "IX" of OMD-LCE-IX stands for implicit exploration, which is justified by the γ_t-biased reward estimator defined as follows. Denote by $Y_{a,a'}^t := \frac{\mathbf{1}[a_n^t = a'] p_a^t q_{a,a'}^t}{p_{a'}^t}(1 - X_n^t)$ the loss of arm a' observed by subroutine a. Let γ_t be a non-negative and non-increasing parameter. We then define the γ_t-biased estimated loss for $Y_{a,a'}^t$ as follows:

$$\hat{Y}_{a,a'}^t := \frac{Y_{a,a'}^t}{q_{a,a'}^t + \gamma_t}.$$

This bias factor γ_t is introduced in [13], [14], which is used to smooth the meta-distributions so that the arms with high loss in the past can still be chosen occasionally for exploration.

In the following, we begin by presenting the fundamental concepts of OMD with Tsallis entropy. We then demonstrate how this approach extends the algorithm studied in our preliminary study [1]. Finally, we explain the efficient implementation of OMD with Tsallis entropy as a subroutine within the OMD-LCE-IX algorithm.

The general idea of the basic OMD with Tsallis entropy as the regularizer works as follows. In the first round, OMD with Tsallis entropy selects a distribution P among arms A that minimizes the Tsallis entropy, i.e., $\psi(P) := \frac{1 - \sum_{a \in A} P(a)^\beta}{1-\beta}$ for $\beta \in (0,1)$, as the regularizer function:

$$P^1 = \arg\min_{P \in \Delta(A)} \psi(P), \tag{5}$$

where $\Delta(A)$ is the simplex on arm set A, i.e., the set of all possible distributions over the arm set. Then, in the subsequent rounds, after obtaining the estimated loss vector $\hat{Y}^t$ among arms, the selection distribution is updated as follows:

$$P^{t+1} = \arg\min_{P \in \Delta(A)} \eta \langle P, \hat{Y}^t \rangle + D_\psi(P, P^t), \tag{6}$$

where $\eta > 0$ is the learning rate, and $D_\psi(P,Q) := \psi(P) - \psi(Q) - \langle \nabla\psi(Q), P - Q \rangle$ is the Bregman divergence associated with ψ.

OMD-LCE-IX is considered as a general version of LCE-IX studied in our preliminary version [1]. The subroutine algorithm in [1] uses the Shannon entropy as the regularizer. On the other hand, the Tsallis entropy can be seen as a generalization of the Shannon entropy. As β approaches 1, the expression $\frac{1 - \sum_{a \in A} P(a)^\beta}{1-\beta}$ converges to $\sum_{a \in A} P(a) \ln(P(a))$.

Next, we show how to adapt the OMD with Tsallis entropy to the swap-regret-minimizing framework, as shown in Alg. 1. For each agent n, OMD-LCE-IX calls OMD with Tsallis entropy $\psi_a(Q) := \frac{1 - \sum_{a' \in A_n} Q(a')^\beta}{1-\beta}$ as each subroutine $a \in A_n$ for any $\beta \in (0,1)$ and distribution $Q \in \Delta(A_n)$. At the very beginning, each subroutine calculates a meta-distribution Q_a^1 according to (5), i.e.,

$$Q_a^1 = \arg\min_{Q \in \Delta(A_n)} \psi_a(Q).$$

By using the method of Lagrange multipliers, each entry $q_{a,a'}^1$ of Q_a^1 can be solved as $q_{a,a'}^1 = \frac{1}{K_n}$. Then, OMD-LCE-IX first calculates P_n^t based on (4) and constructs a γ_t-biased loss estimator based on the played arm and its reward.

Algorithm 1 The OMD-LCE-IX algorithm

1: **Input:** n, A_n, η, γ
2: // Initialization
3: Set $q^t_{a,a'} = \frac{1}{K_n}$ and $\hat{L}^0_{a,a'} = 0, \forall a, a' \in A_n$
4: **for** $t = 1, \ldots, T$ **do**
5:     // Compute the sample distribution, play arms and observe rewards
6:     Calculate P^t_n based on (4)
7:     Play an arm $a^t_n \sim P^t_n$
8:     Observe reward X^t_n
9:     // Update each meta-distribution
10:     **for** $a \in A_n$ **do**
11:         $Y^t_{a,a'} := \frac{\mathbf{1}[a^t_n = a'] p^t_a q^t_{a,a'}}{p^t_{a'}}(1 - X^t_n), \forall a' \in A_n$
12:         $\hat{Y}^t_{a,a'} := \frac{Y^t_{a,a'}}{q^t_{a,a'} + \gamma_t}, \forall a' \in A_n$
13:         Calculate $\tilde{Q}^{t+1}_a$ in the dual space according to

$$\frac{1}{(\tilde{q}^{t+1}_{a,a'})^{1-\beta}} = \frac{1}{(q^t_{a,a'})^{1-\beta}} + \frac{1-\beta}{\beta}\eta \hat{Y}^t_{a,a'}$$

14:         Calculate Q^{t+1}_a by projecting $\tilde{Q}^{t+1}_a$ back to the primal space according to

$$\frac{1}{(q^{t+1}_{a,a'})^{1-\beta}} = \frac{1}{(\tilde{q}^{t+1}_{a,a'})^{1-\beta}} + \lambda^t_a$$

15:     **end for**
16: **end for**

To efficiently solve (6) for each subroutine in the following rounds, invoking Theorem 26.15 in [6], we can obtain a two-step process to obtaining Q^{t+1}_a in the subsequent rounds for each subroutine:

$$\tilde{Q}^{t+1}_a = \arg \min_{Q \in \mathrm{Domain}(\psi_a)} \eta \left\langle Q, \hat{Y}^t_a \right\rangle + D_{\psi_a}\left(Q, Q^t_a\right)$$
$$Q^{t+1}_a = \arg \min_{Q \in \Delta(A_n)} D_{\psi_a}\left(Q, \tilde{Q}^{t+1}_a\right),$$

where $\hat{Y}^t_a$ is the vector of $\hat{Y}^t_{a,a'}$ for all $a' \in A_n$.

The idea of the above two-step update rule is first to calculate a solution $\tilde{Q}^{t+1}_a$ in the dual space (i.e., the domain of the Tsallis entropy), and then project back to the primal space (i.e., $\Delta(A_n)$) by using the Bregman divergence. Notice that the solution to the first step is the solution of the following equation:

$$\eta \hat{Y}^t_{a,a'} + \nabla \psi_a\left(\tilde{Q}^{t+1}_a\right) - \nabla \psi_a\left(Q^t_a\right) = 0. \tag{7}$$

Similarly, we can evaluate the second step explicitly, and the two-step update rule becomes the solutions for the following two equations:

$$\frac{1}{(\tilde{q}^{t+1}_{a,a'})^{1-\beta}} = \frac{1}{(q^t_{a,a'})^{1-\beta}} + \frac{1-\beta}{\beta}\eta \hat{Y}^t_{a,a'},$$
$$\frac{1}{(q^{t+1}_{a,a'})^{1-\beta}} = \frac{1}{(\tilde{q}^{t+1}_{a,a'})^{1-\beta}} + \lambda^t_a,$$

where λ^t_a is a constant from the method of Lagrange multipliers that makes $Q^{t+1}_a := \{\tilde{q}^{t+1}_{a,a'} : \forall a' \in A_n\}$ a distribution, i.e., $\sum_{a' \in A_n} q^{t+1}_{a,a'} = 1$, as shown in Lines 13 to 14. The possible values of γ_t, β and η can be found in Theorem V.3, and the value of λ^t_a can be computed efficiently by a binary search or Newton methods.

V. Analytical Results for OMD-LCE-IX

A. Regret Bounds

Notations. As the regret analysis is for each individual agent $n \in \mathcal{N}$, without confusion, we drop the subscript n in some notations for brevity. Let $\mathcal{G}_t$ denote the σ-algebra generated by the history information of all agents up to round t, i.e., $\mathcal{G}_t := \sigma\left(\{a^1_n, r^1_n, \ldots, a^t_n, r^t_n\}_{n \in \mathcal{N}}\right)$.

We first state a novel concentration bound for the γ_t-biased loss estimator, which directly compares the cumulative estimated swapped loss with the realized swapped loss. In the following, we will slightly abuse the notation β with subscripts and superscripts to represent a random variable. In cases where no subscripts or superscripts are associated with β, it refers to the parameter utilized in the OMD-LCE-IX algorithm.

Lemma V.1. *Let $\delta \in (0,1)$ and let $\beta^t_{a,a'}$ be nonnegative $\mathcal{G}_{t-1}$-measurable random variables (i.e., given $\mathcal{G}_{t-1}$, $\beta^t_{a,a'}$ is determined) satisfying $\beta^t_{a,a'} \le 2\gamma_t$ for all pairs $a, a' \in A_n$. Define $\tilde{Y}^t_{a,a'} := \mathbf{1}[a^t_n = a]y^t_{a'}$. If deterministic numbers $b_t \ge 0$ satisfy $\sum_{a' \in A_n} \beta^t_{a,a'} \le b_t$ for all $a \in A_n$ and $t \in [T]$, then with probability at least $1-\delta$, we have the following inequality:*

$$\sum_{t=1}^{T} \sum_{a \in A_n} \sum_{a' \in A_n} \beta^t_{a,a'} \left(\hat{Y}^t_{a,a'} - \tilde{Y}^t_{a,a'}\right) \le \frac{1}{2} \sum_{t=1}^{T} b_t^2 + \ln \frac{1}{\delta} \ . \tag{8}$$

Proof Sketch. We only give a proof sketch here; the detailed proof can be found in Appendix A. Let

$$X_t := \sum_{a \in A_n} \sum_{a' \in A_n} \beta^t_{a,a'} (\hat{Y}^t_{a,a'} - \tilde{Y}^t_{a,a'}).$$

Conditioning on the other agents' actions $\mathbb{A}^t_{-n}$, we show that $\mathbf{E}_{t-1}[\exp\{X_t\} \mid \mathbb{A}^t_{-n}] \le \exp\{b_t^2/2\}$. Thus,

$$\left\{ \exp\left(\sum_{s=1}^{t} X_s - \frac{1}{2} \sum_{s=1}^{t} b_s^2 \right) \right\}_{t \ge 0}$$

is a supermartingale with respect to $\{\mathcal{G}_t\}_{t\ge 0}$. The result follows by Markov's inequality. □

The proof for Lemma V.1 is refined beyond the IX concentration bounds studied in [14]. The original approach used in [14] is for external regret. However, we cannot simply adapt their concentration bound to analyze the instantaneous swap regret for the following reasons. First, while the swap regret can only be represented as the aggregate of external regrets of subroutine algorithms *in expectation*, this representation does not hold for the instantaneous swap regret. Second, in the original concentration bound, the probability is taken with respect to only one agent's randomness, which is unsuitable for the MAB-UG setting, as the reward/loss for each agent is dependent on all agents' actions. To address the issue, Lemma V.1 considers the K_n subroutines as a whole, and proves concentration between the sum of IX loss estimators for each meta-distribution and the realized swapped losses with respect to all agents' randomness. The following lemma is a direct result of Lemma V.1, which is essential for the swap-regret analysis.

Lemma V.2. *Let $\delta \in (0,1)$. Each of the following inequalities holds with probability at least $1-\delta$:*

$$\sum_{t=1}^{T} \sum_{a \in A_n} \sum_{a' \in A_n} \gamma_t \left(\hat{Y}^t_{a,a'} - \tilde{Y}^t_{a,a'}\right) \le \frac{K_n^2}{2} \sum_{t=1}^{T} \gamma_t^2 + \ln \frac{1}{\delta}, \tag{9}$$

$$\sum_{t=1}^{T} \sum_{a \in A_n} \sum_{a' \in A_n} (q^t_{a,a'})^{1-\beta} \left(\hat{Y}^t_{a,a'} - \tilde{Y}^t_{a,a'}\right) \le \frac{T\gamma_T K_n^{2\beta}}{2} + \frac{1}{\gamma_T} \ln \frac{1}{\delta}, \tag{10}$$

and the following inequality holds simultaneously for all $F \in \mathcal{F}$:

$$\sum_{t=1}^{T} \sum_{a \in A_n} \left(\hat{Y}^t_{a,F(a)} - \mathbf{1}[a^t_n = a] y^t_{F(a)}\right) \le \frac{T\gamma_T}{2} + \frac{K_n}{\gamma_T} \ln \frac{K_n}{\delta}. \tag{11}$$

Proof. (9) is obtained by invoking Lemma V.1 with $\beta^t_{a,a'} = \gamma_t$ and $b_t = K_n\gamma_t$. (10) is obtained by invoking Lemma V.1 with $\beta^t_{a,a'} := \gamma_T (q^t_{a,a'})^{1-\beta}$ for all $a, a' \in A_n$, using $\gamma_T \le \gamma_t$ and $\sum_{a' \in A_n} (q^t_{a,a'})^{1-\beta} \le K_n^{\beta}$. For any fixed F, applying Lemma V.1 with $\beta^t_{a,a'} = \gamma_T \mathbf{1}[a' = F(a)]$ and $b_t = \gamma_T$ gives $\sum_{t=1}^{T} \sum_{a \in A_n} (\hat{Y}^t_{a,F(a)} - \mathbf{1}[a^t_n = a] y^t_{F(a)}) \le T\gamma_T/2 + \gamma_T^{-1} \ln(1/\delta_F)$ with probability at least $1 - \delta_F$. Since $|\mathcal{F}| = K_n^{K_n}$, taking $\delta_F = \delta/|\mathcal{F}|$ and applying a union bound over all swap functions gives (11). The displayed bound in (11) follows from $\ln(K_n^{K_n}/\delta) \le K_n \ln(K_n/\delta)$. □

The regret defined in (3) for each agent $n \in \mathcal{N}$ playing the OMD-LCE-IX algorithm is guaranteed by the following theorem.

Theorem V.3. *Let $\delta \in (0,1)$ and $L_\delta := \ln(8K_n/\delta)$. With probability at least $1-\delta$, $\gamma_t = \sqrt{\frac{L_\delta}{T}}, \forall t \in [T]$, $\beta = \frac{1}{2}$, and $\eta = \sqrt{\frac{K_n}{T}}$, the instantaneous swap regret for playing the OMD-LCE-IX algorithm over T rounds is bounded as follows*

$$R^{\mathrm{swa}}_n(T, \mathcal{F}) \le \left(2K_n + \frac{1}{2}\right)\sqrt{TL_\delta} + 4K_n\sqrt{T} + \frac{K_n^2 L_\delta}{2} + K_n\sqrt{K_n L_\delta} + \left(1 + 2\sqrt{\frac{K_n}{L_\delta}}\right) \ln \frac{8}{\delta}. \tag{12}$$

In particular, $R_n^{\text{swa}}(T,\mathcal{F}) = O(K_n\sqrt{T\ln(K_n/\delta)} + K_n^2\ln(K_n/\delta))$.

Proof Sketch. The regret defined in (3) can be rewritten in the loss form and can be decomposed as follows:

$$R_n^{\text{swap}}(T,\mathcal{F}) \le \max_{F\in\mathcal{F}} \underbrace{\sum_{a\in A_n}(L_a^T - \hat{L}_a^T)}_{=:(\text{a})} + \underbrace{\sum_{a\in A_n}(\hat{L}_a^T - \hat{L}_{a,F(a)}^T)}_{=:(\text{b})} + \underbrace{\sum_{a\in A_n}(\hat{L}_{a,F(a)}^T - \tilde{L}_{a,F(a)}^T)}_{=:(\text{c})},$$

where $L_a^T := \sum_{t=1}^T \sum_{a'\in A_n} Y_{a,a'}^t$, $\hat{L}_a^T := \sum_{t=1}^T\sum_{a'\in A_n} q_{a,a'}^t \hat{Y}_{a,a'}^t$, $\hat{L}_{a,F(a)}^T := \sum_{t=1}^T \hat{Y}_{a,F(a)}^t$, and $\tilde{L}_{a,F(a)}^T := \sum_{t=1}^T \mathbf{1}[a_n^t = a] y_{F(a)}^t$. Then, the proof follows the following four steps.

In Step 1, we bound (a) by $\sum_{a\in A_n}\sum_{t=1}^T \gamma_t \sum_{a'\in A_n} \hat{Y}_{a,a'}^t$, which is a straightforward result by the definition of $\hat{Y}_{a,a'}^t$. In Step 2, we bound (b) by $K_n\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} + \frac{\eta}{\beta}\sum_{t=1}^T\sum_{a\in A_n}\sum_{a'\in A_n}(q_{a,a'}^t)^{1-\beta}\hat{Y}_{a,a'}^t$ by a refined analysis for the online mirror descent algorithm with Tsallis entropy.

In Step 3, invoking Lemma V.2 and taking the union bound over the needed concentration events can upper bound (a), (b), and the realized swapped-loss term (c) with high probability. Thus, with probability at least $1-\delta'$ for $\delta' \in (0,1)$,

$$\begin{aligned}(\text{a}) + (\text{b}) + (\text{c}) \le& \gamma K_n T + \frac{TK_n^2\gamma^2}{2} + K_n\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} + \frac{\eta}{\beta}T(K_n)^\beta \\ &+ \frac{\eta}{\beta}\frac{T\gamma K_n^{2\beta}}{2} + (1+\frac{\eta}{\beta\gamma})\ln\frac{4}{\delta'} + \frac{K_n}{\gamma}\ln\frac{4K_n}{\delta'} + \frac{T\gamma}{2}.\end{aligned}$$

In Step 4, let $\gamma_t = \sqrt{\frac{\ln(8K_n/\delta)}{T}}, \forall t\in[T]$, $\beta = \frac{1}{2}$, and $\eta = \sqrt{\frac{K_n}{T}}$, and loosen the logarithmic constants to obtain (12). The detailed proof can be found in Appendix B. $\square$

Note that it is not required for all agents to play OMD-LCE-IX at the same time to guarantee Theorem V.3. As stated in the following corollary, a confidence-independent choice of γ_t directly leads to the expected swap-regret bound.

Corollary V.4. *With* $\gamma_t = \sqrt{\frac{\ln(8K_n)}{T}}, \forall t\in[T]$, $\beta=\frac{1}{2}$, *and* $\eta = \sqrt{\frac{K_n}{T}}$, *the expected swap regret of OMD-LCE-IX is bounded as follows:*

$$\mathbf{E}[R_n^{\text{swap}}(T,\mathcal{F})] = O(K_n\sqrt{T\ln K_n} + K_n^2\ln K_n).$$

Proof. Let $L_0 := \ln(8K_n)$ and use the confidence-independent parameter $\gamma_t = \sqrt{L_0/T}$ in the high-probability inequality obtained before optimizing γ_t in the proof of Theorem V.3. For every $\delta\in(0,1)$, that inequality has the form

$$R_n^{\text{swap}}(T,\mathcal{F}) \le C_1K_n\sqrt{TL_0} + C_2K_n^2L_0 + C_3K_n\sqrt{K_nL_0} + C_4K_n\sqrt{\frac{T}{L_0}}\ln\frac{1}{\delta}$$

with universal constants $C_1, C_2, C_3, C_4 > 0$. Integrating this tail bound gives $\mathbf{E}[R_n^{\text{swap}}(T,\mathcal{F})] = O(K_n\sqrt{TL_0} + K_n^2L_0) = O(K_n\sqrt{T\ln K_n} + K_n^2\ln K_n)$. $\square$

Note that Corollary V.4 gives an expected swap-regret bound of $O(K_n\sqrt{T\ln K_n} + K_n^2\ln K_n)$ with respect to all agents' randomness.

B. Convergence to Correlated Equilibrium

Bounding the instantaneous swap regret has a significant benefit – it allows us to prove that the OMD-LCE-IX algorithm can converge to a set of correlated equilibria.

Theorem V.5. *If every agent* $n\in\mathcal{N}$ *plays the OMD-LCE-IX algorithm for* T *rounds, then the empirical distribution of the joint actions played by all agents* $\hat{\mathbf{P}}^T$ *is an* ϵ*-CE with probability at least* $1-\delta$, *where*

$$\epsilon = O\left(\max_{n\in\mathcal{N}}\left\{K_n\sqrt{\frac{\ln(K_nN/\delta)}{T}} + \frac{K_n^2\ln(K_nN/\delta)}{T}\right\}\right).$$

When $T\to\infty$, *the empirical distribution of the joint actions converges to a set of correlated equilibria almost surely.*

Proof. Let $\delta' > 0$. By (12) and the property of swap regret, with probability $1-\delta'$,

$$R_n^{\text{int}}(T) \le O\left(K_n\sqrt{T\ln\frac{K_n}{\delta'}} + K_n^2\ln\frac{K_n}{\delta'}\right)$$

for agent n. By using the union bound over all the N agents and letting $\delta' = \delta/N$, we have that with probability at least $1-\delta$,

$$\max_{a,a' \in A_n, \forall n \in \mathcal{N}} \sum_{\mathbb{A}: a_n = a} \hat{\mathbf{P}}^T(\mathbb{A}) \left(r_n\left(a'; \mathbb{A}_{-n}\right) - r_n\left(\mathbb{A}\right)\right)$$
$$\leq \max_{n \in \mathcal{N}} \frac{1}{T} R_n^{\mathrm{int}}(T) \leq O\left(\max_{n \in \mathcal{N}} \left\{K_n \sqrt{\frac{\ln(K_n N/\delta)}{T}} + \frac{K_n^2 \ln(K_n N/\delta)}{T}\right\}\right).$$

To prove the almost-sure convergence to CE, let $\delta_T = 1/T^2$ and denote by E_T the failure event of the above high-probability bound with $\delta = \delta_T$. Then $\sum_{T=1}^{\infty} \Pr(E_T) < \infty$, and the first Borel-Cantelli Lemma gives $\Pr(\limsup_{T\to\infty} E_T) = 0$. Therefore, with probability one, for all sufficiently large T,

$$\max_{a,a' \in A_n, \forall n \in \mathcal{N}} \sum_{\mathbb{A}: a_n = a} \hat{\mathbf{P}}^T(\mathbb{A}) \left(r_n(a'; \mathbb{A}_{-n}) - r_n(\mathbb{A})\right) \leq O\left(\max_{n \in \mathcal{N}} \left\{K_n \sqrt{\frac{\ln(K_n N T^2)}{T}} + \frac{K_n^2 \ln(K_n N T^2)}{T}\right\}\right),$$

whose right-hand side converges to zero. Thus, the empirical distribution of joint actions converges to the set of correlated equilibria almost surely. $\square$

Solving the above bound for T implies that, for $\epsilon \in (0,1)$, the empirical joint distribution $\hat{\mathbf{P}}^T$ for all agents meets the definition of an ϵ-CE for the unknown games after $T = \Omega(\max_{n \in \mathcal{N}} \frac{K_n^2 \ln(K_n N/\delta)}{\epsilon^2})$ rounds, i.e., the ϵ-CE is achieved. Notably, every agent can run the OMD-LCE-IX algorithm with different Tsallis parameters, provided that the instantaneous internal regret is bounded with a high probability. The choice of different β values can influence the rate at which the time-averaged instantaneous internal regret (i.e., $\frac{1}{T} R_n^{\mathrm{int}}(T)$) vanishes. Moreover, the speed at which agents collectively converge to a CE depends on the agent with the slowest vanishing rate of $\frac{1}{T} R_n^{\mathrm{int}}(T)$.

C. Time and Space Complexity

In each round, each agent needs first to calculate P_n^t based on (4), which can be regarded as the calculation of a stationary distribution for an ergodic (i.e., aperiodic and irreducible) Markov chain with K_n states defined by Q_n^t. Such a stationary distribution can be precisely computed in $O(K_n^2)$ time [43], and approximately computed in almost linear time [44]. Then, updating each meta-distribution involves finding λ_a^t. Using binary search, the efficient discovery of λ_a^t only takes $O(\ln(m))$ time, where m is the precision of the number of digits. Thus, when the precision is less than $O(\exp(K_n))$, the time complexity of OMD-LCE-IX is still $O(K_n^2)$. Regarding the space complexity, the space complexity is $O(K_n^2)$.

VI. Numerical Experiments

In this section, we compare OMD-LCE-IX with LCE-IX [1] to show the effectiveness of the OMD technique, and we further compare with LCE (i.e., $\gamma_t = 0$ in LCE-IX) to show the effectiveness of the IX technique. We also compare with a recent algorithm with the full-information feedback called BM-Opt-Hedge [37]. The results are the average of 100 independent experiments.

We study a wireless medium access game between two wireless devices (i.e., two agents), where the two wireless devices are *hidden* from each other (i.e., each device cannot observe each other), and trying to access one unknown channel in each round. Each device has two options in each round, wait for the next round (W) or access in the current round (A). If a device chooses action W, it will receive a reward of 0.

If a device chooses to access (A), the device has an energy cost of 0.2. When only one device successfully accesses the medium, then this device will receive a reward of 0.8. If both devices choose action A, then there is a collision and the rewards for both devices are -0.2 due to the wasted transmission energy. The reward matrix (unknown to the agents) is shown in Table II.

We assume that all the devices do not adopt the RTS/CTS mechanism, an oft-used technique to solve the hidden terminal problem, as it will introduce new problems among its control messages [45] and the game model still applies to the RTS/CTS message itself. Thus, it is quite challenging to improve the received rewards (i.e., the successful access to the channel) for both devices in a distributed way, as *both wireless devices are hidden terminals to each other so that they cannot observe the actions and rewards of each other and they do not know the total number of devices.*

Table II: The reward matrix for the medium access game

	W	A
W	$(0, 0)$	$(0, 0.8)$
A	$(0.8, 0)$	$(-0.2, -0.2)$

As the swap regret is a generic performance measure, different swap functions can lead to different regret definitions. In this experiment, we show two metrics that reflect two different regret definitions. The first metric is the time-averaged reward, the gap of which from the optimal actions in hindsight reflects the external regret of an online learning algorithm. The other metric is the convergence to the ϵ-CE. Whether or not an ϵ-CE can be reached reflects whether an online learning algorithm can minimize its internal regret.

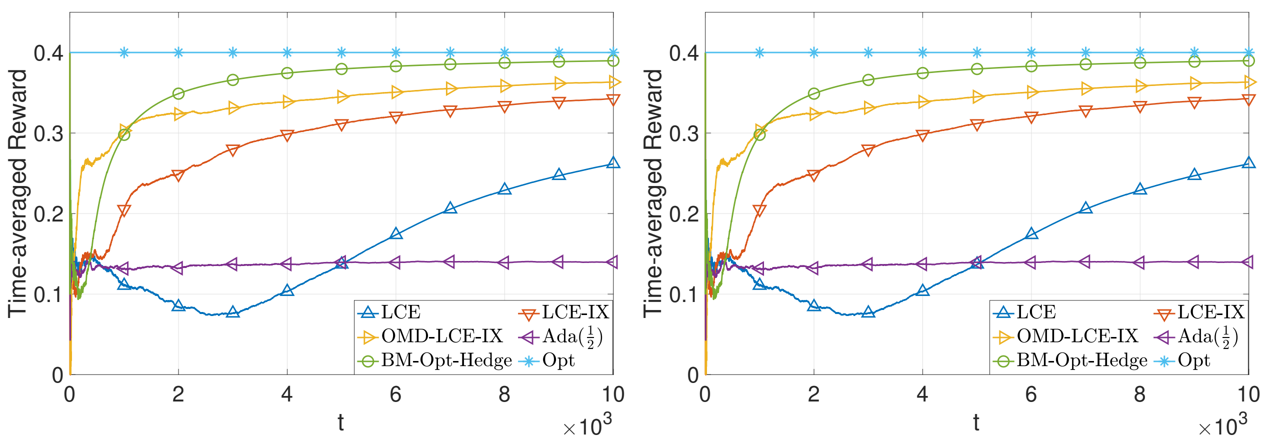
A. Time-averaged Reward

To save space, we only show the time-averaged reward for all the considered algorithms, as it contains the equivalent information about the cumulative regrets or rewards. For example, if an algorithm has higher time-averaged rewards (or closer to the maximum rewards), then it also has higher cumulative rewards (or lower cumulative regrets).

To compare with a benchmark, we consider an adaptive access technique (denoted by `Ada`) with the prior knowledge about the number of all the hidden devices, which randomly accesses a channel with an initial probability $\frac{1}{2}$ for two devices. If a device fails, the access probability of that device is reduced by half; otherwise, the device uses the initial probability in the next round. `Ada` in our experiments can achieve better performance than the distributed coordination function of IEEE 802.11 used by current WiFi devices in real-world scenarios, as `Ada` in our experiments can set an appropriate initial probability to achieve high throughput. Therefore, `Ada` can be a good benchmark with partial prior knowledge to show the effectiveness of the swap-regret-minimizing algorithms.

In addition, the maximum time-averaged reward (denoted by `Opt`) of 0.4 can be achieved, by a mediator (e.g., wireless access point) with full prior knowledge which either asks Agent 1 to play W and Agent 2 to play A, or asks Agent 1 to play A and Agent 2 to play W in each round.

The time-averaged rewards of both agents in 1×10^4 rounds are shown in Fig. 2. As we can see, LCE-IX outperforms both LCE and `Ada`$(\frac{1}{2})$ in terms of the faster convergence to `Opt`. This shows the effectiveness of the γ_t-biased estimator in smoothing the reward estimation so that the low-reward arm can still be explored occasionally. We can also see that OMD-LCE-IX converge faster than LCE-IX, showing the effectiveness of the OMD technique. On the other hand, BM-Opt-Hedge achieves the fastest result, but we note that BM-Opt-Hedge is with the full-information feedback.



(a) Agent 1 (b) Agent 2

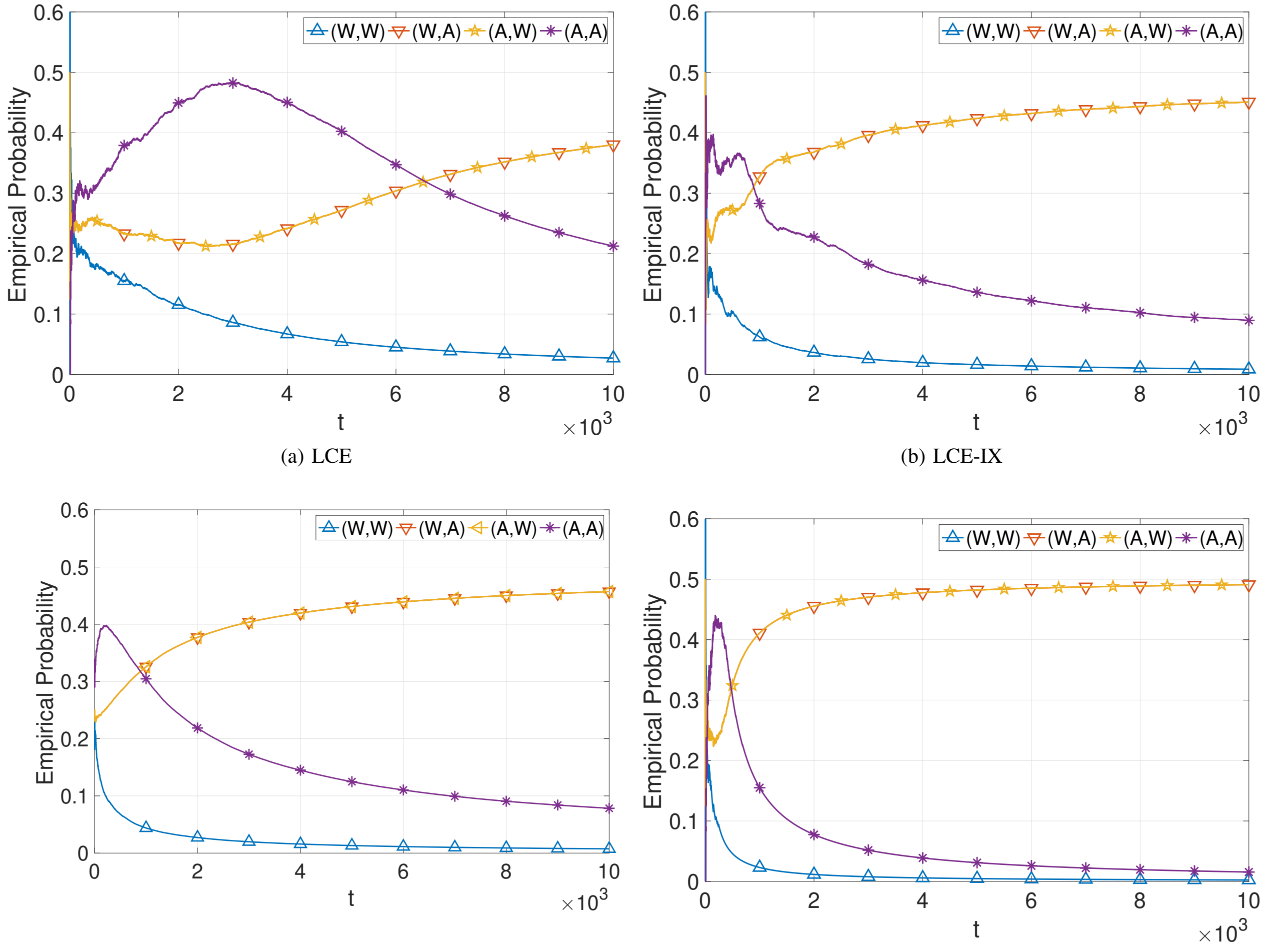
Figure 2: The time-averaged reward for both agents.

B. Convergence to the ϵ-Correlated Equilibrium

The convergence of empirical distribution of joint actions played by the two agents in T rounds is shown in Fig. 3b, where (W,W) means both agents play action W, (W,A) means Agent 1 plays W and Agent 2 plays A, and so on. We take the result of LCE-IX to explain the convergence to the CE. The final results in Fig. 3a are $\hat{\mathbf{P}}^T(\text{W,W}) = 0.0045$, $\hat{\mathbf{P}}^T(\text{W,A}) = 0.4687$, $\hat{\mathbf{P}}^T(\text{A,W}) = 0.4687$, and $\hat{\mathbf{P}}^T(\text{A,A}) = 0.0581$. We can do a simple calculation to verify this empirical distribution is a CE ($\epsilon = 0$). For example, the expected incentives for Agent 1 to switch from W to A are $\hat{\mathbf{P}}^T(\text{W,W}) \cdot u_1(\text{A,W}) + \hat{\mathbf{P}}^T(\text{W,A}) \cdot u_1(\text{A,A}) - (\hat{\mathbf{P}}(\text{W,W}) \cdot u_1(\text{W,W}) + \hat{\mathbf{P}}^T(\text{W,A}) \cdot u_1(\text{W,A})) = -0.0901 < 0$, showing that Agent 1 does not have incentives to switch from W to A when both agents follow the joint distribution $\hat{\mathbf{P}}^T$. In the same way, we can verify the empirical joint distribution $\hat{\mathbf{P}}^T$ is a CE for both agents.

Figs. 3b and 3c show that both OMD-LCE-IX and LCE-IX have a faster convergence than LCE to a CE, as the empirical probabilities of the optimal action pairs of (A,W) and (W,A) increase faster than that of LCE. This again shows the effectiveness of the γ_t-biased estimator in controlling the variation of the reward estimation.

On the other hand, with full-information feedback, BM-Opt-Hedge can achieve a faster convergence rate than OMD-LCE-IX and LCE-IX. It remains to be explored whether the techniques used in BM-Opt-Hedge can be adapted to the bandit-feedback model to accelerate the convergence rate for swap regret.



(a) LCE (b) LCE-IX

(c) OMD-LCE-IX (d) BM-Opt-Hedge

Figure 3: The empirical distribution of joint actions by two agents in T rounds.

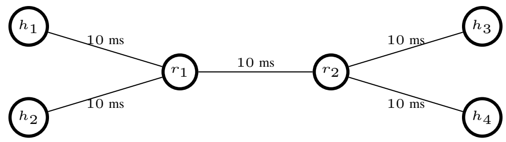
VII. Emulation Experiments

In this section, we apply MAB-UG to the end-to-end congestion control, where the agent in MAB-UG is the data flow, and the action set is the congestion window or sending rates, which are integers as implemented in the Linux kernel. In our framework, a round for each flow is defined as a *round-trip time (RTT)*, which is the interval required for a flow to send packets and receive acknowledgments for these packets. It is important to note that while the minimization of swap-regret does not require a uniform RTT across all agents, our experiments are conducted in topologies (see Fig. 4) where agents have comparably similar RTTs to facilitate convergence to a correlated equilibrium. Additionally, we assume that packet loss is solely due to congestion, mirroring the assumption made by the congestion control mechanism (e.g., CUBIC) in current TCP implementations, which attributes packet loss primarily to congestion.

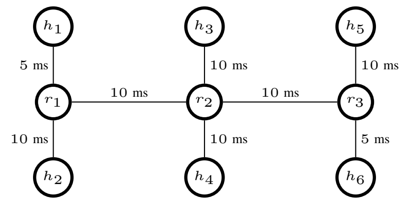
More details about the applications of MAB-UG to the end-to-end congestion control can be found in our paper [2].

We have implemented the proposed OMD-LCE-IX algorithm with the same but normalized utility function defined in [46] through the Linux Kernel 5.13.12 based on the congestion control plane [47], a new API for writing congestion control algorithms. In this section, we start with fairness-related experiments in Mininet [48], where OMD-LCE-IX is compared with CUBIC [49] and BBR version 2 [50] (BBR2 for short in the following). Then, we conduct trace-driven experiments with Pantheon [51], an evaluation platform for congestion control algorithms, with an additional comparison to PCC-Vivace [46]. We have released the source code for the experiments in https://github.com/Zhiming-Huang/OMD-LCE-IX.

In the fairness-related experiments, two classic network topologies recommended by IETF TCP evaluation suite [52] are considered, i.e., the dumbbell and parking lot topologies, as shown in Fig. 4. The bandwidth of all the links in both topologies is 50 Mbps, and the delay of each link is shown in the figures. For both topologies, the queue size on all the links between two routers is 100 packets. In the dumbbell topology, there are two flows from h_1 to h_3 and from h_2 to h_4, sharing the same link r_1-r_2. In the parking lot topology, there are three flows from h_1 to h_6, from h_2 to h_3, and from h_4 to h_5. We can see that the flow from h_1 to h_6 competes with both the other two flows, while the other two flows are independent of each other. In the experiments, we use `iperf` to generate a 30s test for the performance of the three congestion control algorithms. As



(a) The dumbbell topology

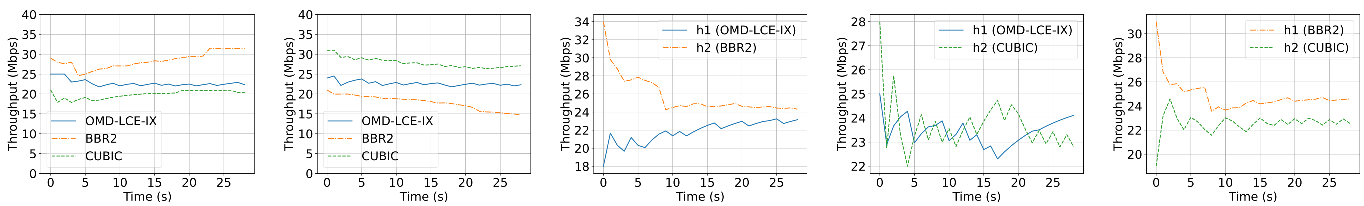


(b) The parking lot topology

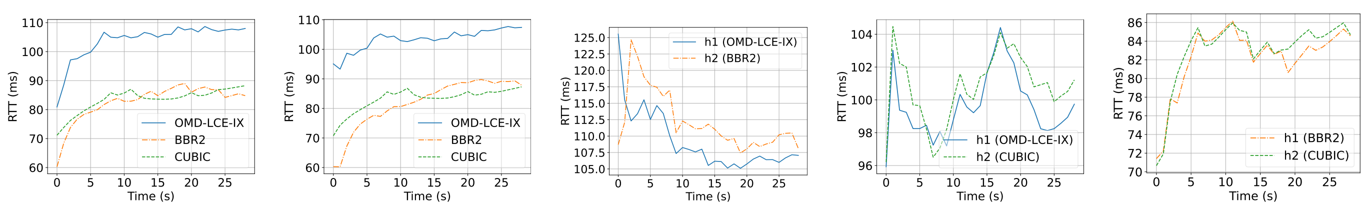
Figure 4: The experiment topology.

`iperf` outputs the averaged results (i.e., throughput and RTT) every 1s, the points at time 0s in Figs. 5 to 7 are the averaged results in the initial interval from 0s to 1s, and then we use the exponentially weighted moving average technique to smooth the results in the following time.

In the trace-driven experiments, we use the US cellular network traces (i.e., T-mobile and Verizon) recorded by the saturator tool [53] while driving. These traces represent the time-varying capacity of the networks experienced by a mobile user, so we can test the adaptability of congestion control algorithms in such network environments.



(a) Homo-flow Throughput on h_1 (b) Homo-flow Throughput on h_2 (c) Throughput (OMD-LCE-IX, BBR2) (d) Throughput (OMD-LCE-IX, CUBIC) (e) Throughput (BBR2, CUBIC)

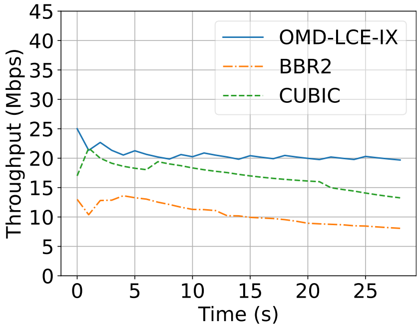


(f) Homo-flow RTT on h_1 (g) Homo-flow RTT on h_2 (h) RTT (OMD-LCE-IX, BBR2) (i) RTT (OMD-LCE-IX, CUBIC) (j) RTT (BBR2, CUBIC)

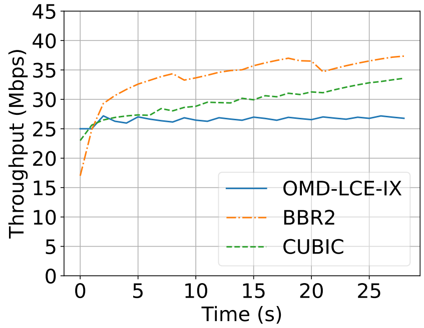
Figure 5: The experiment results for the dumbbell topology.

A. Dumbbell Results

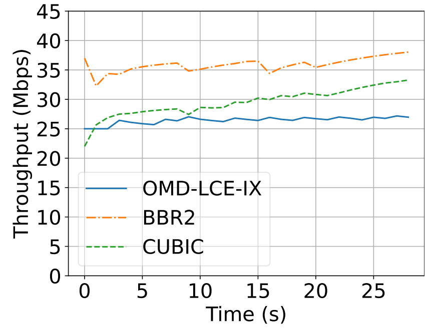
We first test the scenario where the two flows are homogeneous, i.e., both the flows adopt the same congestion control algorithm. The throughput and RTT results for the homogeneous flows are shown in Figs. 5a, 5b, 5f and 5g. As observed, when all flows implement OMD-LCE-IX, they tend to achieve a fair performance (i.e., similar throughput). This is because OMD-LCE-IX ensures convergence to a CE when adopted by all flows, simulating the effect of a central controller that distributes bandwidth evenly among all participating flows. However, the other two congestion control algorithms, i.e., CUBIC and BBR2, do not guarantee an equal share of resources between the flows. This is due to the intrinsic property of these algorithms (i.e., deterministic strategy) and the slight difference in flow start times, although we made efforts to minimize this difference in the experiments by using a script to control all nodes. On the other hand, we can observe OMD-LCE-IX performs better than BBR2 and CUBIC in the homogeneous setting in terms of throughput, while having a higher RTT. This is because OMD-LCE-IX does not explicitly incorporate queuing models like BBR2 does, and its randomized action selection makes it less conservative in utilizing network buffers, leading to increased queue lengths.



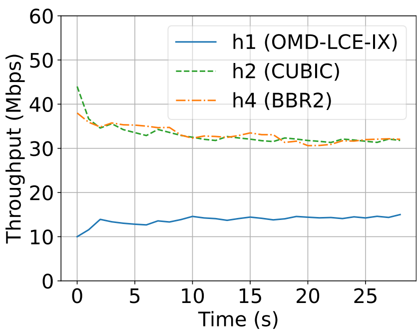
(a) Homo-flow
Throughput on h_1



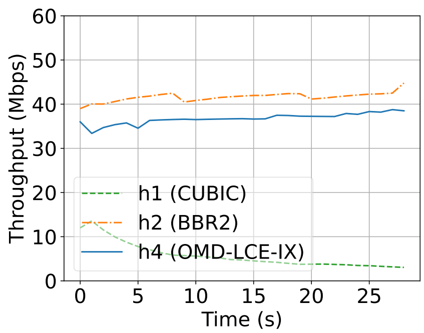
(b) Homo-flow
Throughput on h_2



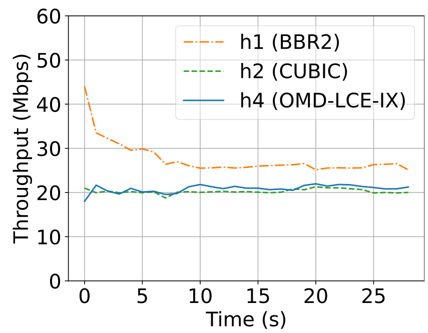
(c) Homo-flow
Throughput on h_4



(d) Throughput for
Hete-flow Setting 1



(e) Throughput for
Hete-flow Setting 2



(f) Throughput for
Hete-flow Setting 3

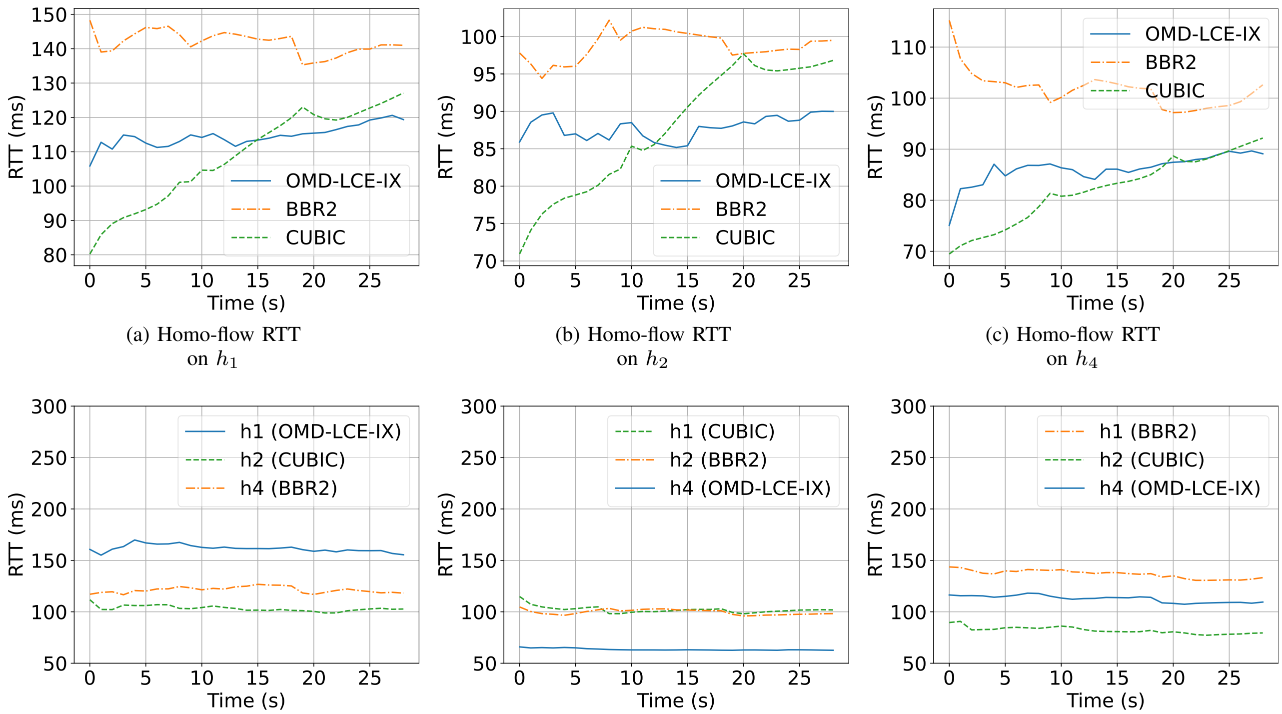
Figure 6: The throughput results for the parking lot topology.

We also conduct experiments for the heterogeneous flow, i.e., the two flows adopt different congestion control algorithms, as shown in Figs. 5c to 5e and Figs. 5h to 5j. When BBR2 and OMD-LCE-IX compete with each other, BBR2 prevails at first but their gap gradually decreases due to the benefits of learning. On the other hand, OMD-LCE-IX can achieve a similar throughput as CUBIC. Regarding RTT, we can observe in Fig. 5h that in the first few intervals, both OMD-LCE-IX and BBR2 suffer a high RTT because they are competing with each other to exhaust the network bandwidth. Overall, the RTTs for both algorithms are decreasing over time, meaning that the network is getting less congested. Therefore, OMD-LCE-IX is friendly to and competitive with other TCP flows.

B. Parking Lot Results

The results of the parking lot topology are shown in Figs. 6 and 7. Similar to the dumbbell topology, we first test the homogeneous flows in the parking lot topology, as shown in Figs. 6a to 6c and Figs. 7a to 7c. The flow from h_1 to h_6 will suffer a loss if any one of the other two flows suffers, i.e., the flow from h_1 to h_6 has a higher probability of suffering a loss and thus results in a lower throughput than the other two flows. As we can see, when all the three flows play the same congestion control algorithm, OMD-LCE-IX always guarantees that the performance gaps between flows are similar and not excessive. On the other hand, CUBIC and BBR2 have a large performance gap between flow h_1 to h_6 and the other two flows. This reflects OMD-LCE-IX's ability to achieve a stable CE. Also, CUBIC and BBR2 incur a higher RTT than OMD-LCE-IX, because CUBIC and BBR2 will increase the *srate* to probe for possible higher bandwidth, while OMD-LCE-IX can maintain a low RTT.

Then, we perform three different heterogeneous flow settings in the parking lot topology for a different 4-hop flow (i.e., flow h_1 to h_6). In the first heterogeneous flow setting, flow h_1 to h_6 adopts OMD-LCE-IX, flow h_2 to h_3 adopts CUBIC, and flow h_4 to h_5 adopts BBR2, and the results are shown in Figs. 6d and 7d. In the second setting, the 4-hop flow h_1 to h_6 adopts CUBIC, flow h_2 to h_3 adopts BBR2 and flow h_4 to h_5 adopts OMD-LCE-IX, as shown in Figs. 6e and 7e. In the third setting, the 4-hop flow h_1 to h_6 would be BBR2, and the other two flows adopt CUBIC and OMD-LCE-IX, respectively, as shown in Figs. 6f and 7f. We can see that when the 4-hop flow adopts OMD-LCE-IX or CUBIC, it will concede the link bandwidth to the other two flows, but OMD-LCE-IX still maintains a higher throughput than CUBIC, as shown in Figs. 6d and 6e. However, when the 4-hop flow adopts BBR2, it occupies a comparable link bandwidth with the other two flows, as shown in Fig. 6f. Overall, from the above experiments, we can see that OMD-LCE-IX is friendly but competitive to other flows, and maintains fairness in allocating the link bandwidth.



(a) Homo-flow RTT on h_1

(b) Homo-flow RTT on h_2

(c) Homo-flow RTT on h_4

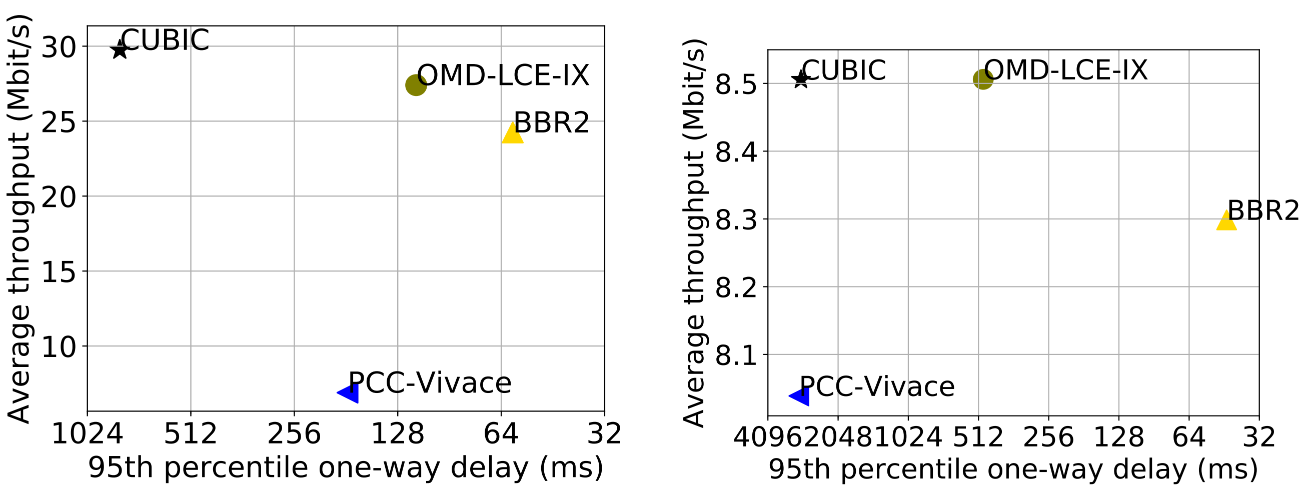
(d) RTT for Hete-flow Setting 1

(e) RTT for Hete-flow Setting 2

(f) RTT for Hete-flow Setting 3

Figure 7: The RTT results for the parking lot topology.

C. Trace-driven Experiments



(a) T-mobile LTE Network

(b) Verizon LTE Network

Figure 8: The trace-driven experiment results on Pantheon.

For each trace, we did ten independent runs of experiments, and the mean results for the trace-driven experiments produced by Pantheon are shown in Fig. 8. Pantheon evaluates an algorithm based on two metrics, i.e., mean throughput and 95th-percentile one-way delay. In both LTE networks, CUBIC achieves the highest throughput albeit at the cost of the highest delay. OMD-LCE-IX, on the other hand, offers a throughout that is comparable to CUBIC but with a lower delay. Conversely, BBR2 achieves the lowest delay but scarifies the throughput which is lower than that of OMD-LCE-IX. Overall, we can see that OMD-LCE-IX can better balance the tradeoff between the average throughput and delay in these two metrics for the T-mobile network than the other three algorithms. Thus, OMD-LCE-IX can guarantee good performance in a dynamic network environment.

VIII. Conclusion

In this paper, we studied the MAB-UG problem and proposed OMD-LCE-IX to address it. With regard to the randomness of all agents' actions, we provided the high-probability bounds for the instantaneous swap regret, which further bound the expected swap regret. Furthermore, we conducted numerical and emulation experiments to show the possible applications of the MAB-UG problems in wireless access control and end-to-end congestion control.

Regarding future work, we aim to close the gap between the upper and lower bounds for swap regret. The minimax-optimal dependence on K_n in the bandit settings remains open, and we will refine the lower bound for swap regret in these settings.

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APPENDIX

As the proofs are for each individual agent n, without confusion, we drop the subscript n in some notations for brevity. Recall that $\mathcal{G}_t$ the σ-algebra generated by the history information of all agents till round t and let $\mathbf{E}_t[\cdot] := \mathbf{E}[\cdot|\mathcal{G}_t]$ be the expectation conditioned on the history information by the end of round t. Recall that $y_a^t := 1 - u_n^t(a; \mathbb{A}_{-n}^t)$ is the instantaneous loss function if agent n plays arm $a \in A_n$ in round t, and thus $Y_{a,a'}^t := \frac{\mathbf{1}[a_n^t = a'] p_a^t q_{a,a'}^t y_{a'}^t}{p_{a'}^t}$ and $\hat{Y}_{a,a'}^t = \frac{Y_{a,a'}^t}{q_{a,a'}^t + \gamma_t}$. Denote by $L_a^T := \sum_{t=1}^T \sum_{a' \in A_n} Y_{a,a'}^t$, $\hat{L}_a^T := \sum_{t=1}^T \sum_{a' \in A_n} q_{a,a'}^t \hat{Y}_{a,a'}^t$, $\hat{L}_{a,a'}^T := \sum_{t=1}^T \hat{Y}_{a,a'}^t$, and $\tilde{L}_{a,a'}^T := \sum_{t=1}^T \mathbf{1}[a_n^t = a] y_{a'}^t$.

A. Proof of Lemma V.1

Proof. Let

$$X_t := \sum_{a \in A_n} \sum_{a' \in A_n} \beta_{a,a'}^t \left(\hat{Y}_{a,a'}^t - \tilde{Y}_{a,a'}^t \right).$$

It suffices to prove $\mathbf{E}_{t-1}[\exp\{X_t\}] \le \exp\{b_t^2/2\}$, because then $\{\exp(\sum_{s=1}^t X_s - \frac{1}{2}\sum_{s=1}^t b_s^2)\}_{t \ge 0}$ is a supermartingale and Markov's inequality gives (8).

Condition on $\mathbb{A}_{-n}^t$, so that y_a^t is fixed, and let j be the action selected by agent n at round t. Define $T_j := \sum_{a' \in A_n} \beta_{j,a'}^t y_{a'}^t$. Then, given $a_n^t = j$,

$$X_t = \sum_{a \in A_n} \beta_{a,j}^t \frac{p_a^t q_{a,j}^t y_j^t}{p_j^t (q_{a,j}^t + \gamma_t)} - T_j.$$

By Jensen's inequality, $\beta_{a,j}^t \le 2\gamma_t$, and $\frac{z}{1+z/2} \le \ln(1+z)$ for $z \ge 0$, we have

$$\begin{aligned}
\exp\left\{ \sum_{a \in A_n} \beta_{a,j}^t \frac{p_a^t q_{a,j}^t y_j^t}{p_j^t (q_{a,j}^t + \gamma_t)} \right\} &\le \sum_{a \in A_n} \frac{p_a^t q_{a,j}^t}{p_j^t} \exp\left\{ \frac{\beta_{a,j}^t y_j^t}{q_{a,j}^t + \gamma_t} \right\} \\
&\le \sum_{a \in A_n} \frac{p_a^t q_{a,j}^t}{p_j^t} \left(1 + \frac{\beta_{a,j}^t y_j^t}{q_{a,j}^t} \right) = 1 + \frac{y_j^t}{p_j^t} \sum_{a \in A_n} p_a^t \beta_{a,j}^t.
\end{aligned}$$

Taking the expectation over a_n^t conditionally on $\mathbb{A}_{-n}^t$ gives

$$\begin{aligned}\mathbf{E}_{t-1}[\exp\{X_t\} \mid \mathbb{A}_{-n}^t] &\le \sum_{j\in A_n} p_j^t e^{-T_j} + \sum_{j\in A_n} y_j^t e^{-T_j} \sum_{a\in A_n} p_a^t \beta_{a,j}^t \\ &\le \sum_{a\in A_n} p_a^t \left(e^{-T_a} + T_a\right) \le \sum_{a\in A_n} p_a^t \left(1+\frac{T_a^2}{2}\right) \le e^{b_t^2/2}.\end{aligned}$$

The second inequality uses $e^{-T_j} \le 1$, and the third one uses $e^{-x} \le 1 - x + x^2/2$ for $x \ge 0$. Since $0 \le T_a \le \sum_{a'} \beta_{a,a'}^t \le b_t$, the last inequality follows. The above bound holds for every realization of $\mathbb{A}_{-n}^t$. Hence, by the tower property,

$$\begin{aligned}\mathbf{E}_{t-1}[\exp\{X_t\}] &= \mathbf{E}_{t-1}\left[\mathbf{E}_{t-1}[\exp\{X_t\} \mid \mathbb{A}_{-n}^t]\right] \\ &\le \mathbf{E}_{t-1}\left[e^{b_t^2/2}\right] = e^{b_t^2/2},\end{aligned}$$

where b_t is deterministic and $\beta_{a,a'}^t$ is $\mathcal{G}_{t-1}$-measurable. Define

$$M_t := \exp\left\{\sum_{s=1}^t X_s - \frac{1}{2}\sum_{s=1}^t b_s^2\right\}, \qquad M_0 := 1.$$

Then,

$$\mathbf{E}_{t-1}[M_t] = M_{t-1} e^{-b_t^2/2} \mathbf{E}_{t-1}[e^{X_t}] \le M_{t-1},$$

so $\{M_t\}_{t\ge 0}$ is a nonnegative supermartingale and $\mathbf{E}[M_T] \le \mathbf{E}[M_0] = 1$. Therefore, by Markov's inequality,

$$\begin{aligned}&\Pr\left(\sum_{t=1}^T X_t > \frac{1}{2}\sum_{t=1}^T b_t^2 + \ln\frac{1}{\delta}\right) \\ &= \Pr\left(M_T > \frac{1}{\delta}\right) \le \delta\, \mathbf{E}[M_T] \le \delta.\end{aligned}$$

Substituting the definition of X_t proves (8). □

B. Proof of Theorem V.3

Proof. By the relationship between P_n^t and Q_a^t, we have the following equation held:

$$\begin{aligned}\sum_{a\in A_n} L_a^T &= \sum_{a\in A_n}\sum_{t=1}^T \sum_{a'\in A_n} Y_{a,a'}^t = \sum_{t=1}^T \sum_{a'\in A_n}\sum_{a\in A_n} \frac{\mathbf{1}[a_n^t = a'] p_a^t q_{a,a'}^t}{p_{a'}^t} y_{a'}^t \\ &= \sum_{t=1}^T \sum_{a'\in A_n} \mathbf{1}[a_n^t = a'] y_{a'}^t = \sum_{t=1}^T \sum_{a\in A_n} \mathbf{1}[a_n^t = a] y_a^t,\end{aligned} \tag{13}$$

The regret defined in (3) can be rewritten in the loss form and can be decomposed as follows:

$$\begin{aligned}&R_n^{\text{swa}}(T,\mathcal{F}) \\ &= \max_{F\in\mathcal{F}} \sum_{t=1}^T \sum_{a\in A_n} \mathbf{1}[a_n^t = a] y_a^t - \sum_{t=1}^T \sum_{a\in A_n} \mathbf{1}[a_n^t = a] y_{F(a)}^t \\ &= \max_{F\in\mathcal{F}} \sum_{a\in A_n} L_a^T - \sum_{a\in A_n} \tilde{L}_{a,F(a)}^T \\ &= \max_{F\in\mathcal{F}} \underbrace{\sum_{a\in A_n} (L_a^T - \hat{L}_a^T)}_{=:(\mathrm{a})} + \underbrace{\sum_{a\in A_n} (\hat{L}_a^T - \hat{L}_{a,F(a)}^T)}_{=:(\mathrm{b})} + \underbrace{\sum_{a\in A_n} (\hat{L}_{a,F(a)}^T - \tilde{L}_{a,F(a)}^T)}_{=:(\mathrm{c})},\end{aligned} \tag{14}$$

where the second equality is due to (13) and the definition of $\tilde{L}_{a,F(a)}^T := \sum_{t=1}^T \mathbf{1}[a_n^t = a] y_{F(a)}^t$. In the following steps, we will give an upper bound to (a), (b), and (c) for any $F \in \mathcal{F}$.

Step 1: Bound (a). By definition of L_a^T and $\hat{L}_a^T$, we have that

$$\begin{aligned}L_a^T - \hat{L}_a^T &= \sum_{t=1}^T \sum_{a'\in A_n} Y_{a,a'}^t - \sum_{t=1}^T \sum_{a'\in A_n} q_{a,a'}^t \hat{Y}_{a,a'}^t \\ &= \sum_{t=1}^T \sum_{a'\in A_n} Y_{a,a'}^t \left(1 - \frac{q_{a,a'}^t}{q_{a,a'}^t + \gamma_t}\right) = \sum_{t=1}^T \gamma_t \sum_{a'\in A_n} \hat{Y}_{a,a'}^t.\end{aligned} \tag{15}$$

Thus, (a) is bounded by $\sum_{t=1}^{T}\gamma_t \sum_{a\in A_n}\sum_{a'\in A_n}\hat{Y}^t_{a,a'}$. To keep concise, we will analyze a fixed value γ_t, which we refer to as γ throughout the following analysis. The results can be readily extended to a variable γ_t over time.

Step 2: Bound (b). By the definition of $\hat{L}^T_a$ and $\hat{L}^T_{a,F(a)}$, we have that

$$\begin{aligned}
\text{(b)} &= \sum_{t=1}^{T}\sum_{a\in A_n}\sum_{a'\in A_n} q^t_{a,a'}\hat{Y}^t_{a,a'} - \sum_{t=1}^{T}\sum_{a\in A_n}\hat{Y}^t_{a,F(a)} \\
&\le \max_{Q\in\Delta(A_n)}\sum_{t=1}^{T}\sum_{a\in A_n}\left(\sum_{a'\in A_n} q^t_{a,a'}\hat{Y}^t_{a,a'} - \sum_{a'\in A_n} Q(a')\hat{Y}^t_{a,a'}\right).
\end{aligned}$$

For any $Q\in\Delta(A_n)$, we have that

$$\begin{aligned}
&\sum_{a'\in A_n} q^t_{a,a'}\hat{Y}^t_{a,a'} - \sum_{a'\in A_n} Q(a')\hat{Y}^t_{a,a'} \\
&= \sum_{a'\in A_n}\left(q^t_{a,a'} - q^{t+1}_{a,a'}\right)\hat{Y}^t_{a,a'} - \sum_{a'\in A_n}\left(q^{t+1}_{a,a'} - Q(a')\right)\hat{Y}^t_{a,a'}.
\end{aligned} \tag{16}$$

Let $\hat{Y}^t_a := [\hat{Y}^t_{a,a'}]_{a'\in A_n}$ denote the vector of loss estimators for meta-distribution a. Then, the first term on the RHS of the above equation can be bounded by using (7) and the definition of Bregman divergence as follows:

$$\begin{aligned}
&\sum_{a'\in A_n}\left(q^t_{a,a'} - q^{t+1}_{a,a'}\right)\hat{Y}^t_a = \left\langle Q^t_a - Q^{t+1}_a, \hat{Y}^t_{a,a'}\right\rangle \\
&= \frac{1}{\eta}\left\langle Q^t_a - Q^{t+1}_a, \nabla\psi_a\left(Q^t_a\right) - \nabla\psi_a\left(\tilde{Q}^{t+1}_a\right)\right\rangle \\
&= \frac{1}{\eta}\left(D_{\psi_a}\left(Q^{t+1}_a, Q^t_a\right) + D_{\psi_a}\left(Q^t_a, \tilde{Q}^{t+1}_a\right) - D_{\psi_a}\left(Q^{t+1}_a, \tilde{Q}^{t+1}_a\right)\right) \\
&\le \frac{1}{\eta}\left(D_{\psi_a}\left(Q^{t+1}_a, Q^t_a\right) + D_{\psi_a}\left(Q^t_a, \tilde{Q}^{t+1}_a\right)\right).
\end{aligned}$$

The second item in the RHS of (16) can be bounded by using the first-order optimality conditions for Q^{t+1}_a and the fact that the Tsallis entropy is convex. By definition, $Q^{t+1}_a := \arg\min_{Q\in\Delta(A_n)}\eta\left\langle Q, \hat{Y}^t_a\right\rangle + D_{\psi_a}(Q, Q^t_a)$, and the first-order optimality conditions for Q^{t+1}_a shows that for any $Q\in\Delta(A_n)$: $\left\langle Q - Q^{t+1}_a, \eta\hat{Y}^t_{a,a'} + \nabla\psi_a\left(Q^{t+1}_a\right) - \nabla\psi_a\left(Q^t_a\right)\right\rangle \ge 0$, which indicates that the second term on the RHS of (16) can be bounded as follows:

$$\begin{aligned}
&\sum_{a'\in A_n}\left(q^{t+1}_{a,a'} - Q(a')\right)\hat{Y}^t_{a,a'} = \left\langle Q^{t+1}_a - Q, \hat{Y}^t_a\right\rangle \\
&\le \frac{1}{\eta}\left\langle Q - Q^{t+1}_a, \nabla\psi_a(Q^{t+1}_a) - \nabla\psi_a(Q^t_a)\right\rangle \\
&= \frac{1}{\eta}\left(D_{\psi_a}\left(Q, Q^t_a\right) - D_{\psi_a}\left(Q, Q^{t+1}_a\right) - D_{\psi_a}\left(Q^{t+1}_a, Q^t_a\right)\right)
\end{aligned}$$

Thus, plugging the above two inequalities into (16) gives

$$\begin{aligned}
&\sum_{t=1}^{T}\left(\sum_{a'\in A_n} q^t_{a,a'}\hat{Y}^t_{a,a'} - \sum_{a'\in A_n} Q(a')\hat{Y}^t_{a,a'}\right) \\
&\le \frac{1}{\eta}\sum_{t=1}^{T}\left(D_{\psi_a}(Q, Q^t_a) - D_{\psi_a}(Q, Q^{t+1}_a) + D_{\psi_a}(Q^t_a, \tilde{Q}^{t+1}_a)\right) \\
&\le \frac{D_{\psi_a}(Q, Q^1_a)}{\eta} + \frac{1}{\eta}\sum_{t=1}^{T} D_{\psi_a}(Q^t_a, \tilde{Q}^{t+1}_a).
\end{aligned}$$

When Q concentrates on a particular action, $D_{\psi_a}(Q, Q_a^1) = \frac{(K_n)^{1-\beta}-1}{1-\beta}$. Thus, summing the above inequality over $a \in A_n$, we can bound (b) as follows:

$$
\begin{aligned}
\text{(b)} &\leq \sum_{a\in A_n}\left(\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} + \frac{1}{\eta}\sum_{t=1}^{T} D_{\psi_a}(Q_a^t, \tilde{Q}_a^{t+1})\right) \\
&= \sum_{a\in A_n}\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} + \frac{1}{\eta}\sum_{a\in A_n}\sum_{t=1}^{T}\frac{1}{1-\beta} \\
&\quad \cdot \sum_{a'\in A_n}\left((\tilde{q}_{a,a'}^{t+1})^{\beta} - (q_{a,a'}^{t})^{\beta} + \frac{\beta}{(\tilde{q}_{a,a'}^{t+1})^{1-\beta}}\left(q_{a,a'}^{t} - \tilde{q}_{a,a'}^{t+1}\right)\right),
\end{aligned}
$$

where the last equality is due to the definitions of the Bregman divergence and the Tsallis entropy. Plugging the first equation from the two-step update rules, i.e., $\frac{1}{(\tilde{q}_{a,a'}^{t+1})^{1-\beta}} = \frac{1}{(q_{a,a'}^{t})^{1-\beta}} + \frac{1-\beta}{\beta}\eta\hat{Y}_{a,a'}^{t}$, into the above function gives

$$
\begin{aligned}
\text{(b)} &\leq \sum_{a\in A_n}\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} \\
&\quad + \frac{1}{\eta}\sum_{a\in A_n}\sum_{t=1}^{T}\sum_{a'\in A_n}\left((\tilde{q}_{a,a'}^{t+1})^{\beta} - (q_{a,a'}^{t})^{\beta} + \eta q_{a,a'}^{t}\hat{Y}_{a,a'}^{t}\right).
\end{aligned} \tag{17}
$$

Notice that

$$
\begin{aligned}
(\tilde{q}_{a,a'}^{t+1})^{\beta} &= (q_{a,a'}^{t})^{\beta}\left(\frac{\tilde{q}_{a,a'}^{t+1}}{q_{a,a'}^{t}}\right)^{\beta} = (q_{a,a'}^{t})^{\beta}\left(\frac{(q_{a,a'}^{t})^{1-\beta}}{(\tilde{q}_{a,a'}^{t+1})^{1-\beta}}\right)^{\frac{\beta}{\beta-1}} \\
&= (q_{a,a'}^{t})^{\beta}\left(1 + \frac{1-\beta}{\beta}\eta(q_{a,a'}^{t})^{1-\beta}\hat{Y}_{a,a'}^{t}\right)^{\frac{\beta}{\beta-1}} \\
&\leq (q_{a,a'}^{t})^{\beta}\left(1 - \eta(q_{a,a'}^{t})^{1-\beta}\hat{Y}_{a,a'}^{t} + \frac{\eta^2}{\beta}(q_{a,a'}^{t})^{2-2\beta}(\hat{Y}_{a,a'}^{t})^2\right) \\
&= \left((q_{a,a'}^{t})^{\beta} - \eta q_{a,a'}^{t}\hat{Y}_{a,a'}^{t} + \frac{\eta^2}{\beta}(q_{a,a'}^{t})^{2-\beta}(\hat{Y}_{a,a'}^{t})^2\right)
\end{aligned}
$$

where the third equality is again due to $\frac{1}{(\tilde{q}_{a,a'}^{t+1})^{1-\beta}} = \frac{1}{(q_{a,a'}^{t})^{1-\beta}} + \frac{1-\beta}{\beta}\eta\hat{Y}_{a,a'}^{t}$, and the inequality is due to that $(1+x)^{\alpha} \leq 1 + \alpha x + \alpha(\alpha-1)x^2$ for any $x \geq 0$ and $\alpha < 0$. Plugging the above inequality into (17) gives

$$
\begin{aligned}
\text{(b)} &\leq \sum_{a\in A_n}\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} + \frac{\eta}{\beta}\sum_{t=1}^{T}\sum_{a\in A_n}\sum_{a'\in A_n}(q_{a,a'}^{t})^{2-\beta}(\hat{Y}_{a,a'}^{t})^2 \\
&\leq K_n\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} + \frac{\eta}{\beta}\sum_{t=1}^{T}\sum_{a\in A_n}\sum_{a'\in A_n}(q_{a,a'}^{t})^{1-\beta}\hat{Y}_{a,a'}^{t}
\end{aligned}
$$

where the second inequality is due to the fact $q_{a,a'}^{t}\hat{Y}_{a,a'}^{t} \leq 1$.

Step 3: Bound (a) + (b) + (c). Combining (a), (b) and (c), and invoking Lemma V.2 with a union bound over the required concentration events guarantees the following inequality with probability at least $1-\delta'$:

$$
\begin{aligned}
\text{(a)} + \text{(b)} + \text{(c)} &\leq \gamma\sum_{t=1}^{T}\sum_{a\in A_n}\sum_{a'\in A_n}\tilde{Y}_{a,a'}^{t} + \frac{TK_n^2\gamma^2}{2} + K_n\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} \\
&\quad + \frac{\eta}{\beta}\sum_{t=1}^{T}\sum_{a\in A_n}\sum_{a'\in A_n}(q_{a,a'}^{t})^{1-\beta}\tilde{Y}_{a,a'}^{t} + \frac{\eta}{\beta}\frac{T\gamma K_n^{2\beta}}{2} \\
&\quad + (1+\frac{\eta}{\beta\gamma})\ln\frac{4}{\delta'} + \frac{K_n}{\gamma}\ln\frac{4K_n}{\delta'} + \frac{T\gamma}{2} \\
&\leq \gamma K_n T + \frac{TK_n^2\gamma^2}{2} + K_n\frac{(K_n)^{1-\beta}-1}{(1-\beta)\eta} + \frac{\eta}{\beta}T(K_n)^{\beta} \\
&\quad + \frac{\eta}{\beta}\frac{T\gamma K_n^{2\beta}}{2} + (1+\frac{\eta}{\beta\gamma})\ln\frac{4}{\delta'} + \frac{K_n}{\gamma}\ln\frac{4K_n}{\delta'} + \frac{T\gamma}{2}.
\end{aligned}
$$

where the first inequality is due to Lemma V.2 to bound the direct realized estimator gaps and the realized swapped-loss term (c) uniformly over $\mathcal{F}$. The second inequality uses $\sum_{a,a'}\hat{Y}_{a,a'}^{t} \leq K_n$ and Hölder's inequality, $\sum_{a'\in A_n}(q_{a_n^t,a'}^{t})^{1-\beta}y_{a'}^{t} \leq K_n^{\beta}$.

Step 4: Substitute parameters. Applying Step 3 with $\delta' = \delta$ and loosening the logarithmic constants gives, with probability at least $1 - \delta$,

$$\begin{aligned} R_n^{\mathrm{swa}}(T, \mathcal{F}) \leq & \gamma K_n T + \frac{T K_n^2 \gamma^2}{2} + K_n \frac{(K_n)^{1-\beta} - 1}{(1-\beta)\eta} + \frac{\eta}{\beta} T (K_n)^{\beta} \\ & + \frac{\eta}{\beta} \frac{T \gamma K_n^{2\beta}}{2} + (1 + \frac{\eta}{\beta\gamma}) \ln(\frac{8}{\delta}) + \frac{K_n}{\gamma} \ln(\frac{8K_n}{\delta}) + \frac{T\gamma}{2}. \end{aligned} \tag{18}$$

The theorem follows by letting $L_\delta = \ln(8K_n/\delta)$, $\gamma = \sqrt{\frac{L_\delta}{T}}$, $\beta = \frac{1}{2}$, and $\eta = \sqrt{\frac{K_n}{T}}$. The above inequality becomes

$$R_n^{\mathrm{swa}}(T, \mathcal{F}) \leq \left(2K_n + \frac{1}{2}\right) \sqrt{T L_\delta} + 4K_n \sqrt{T} + \frac{K_n^2 L_\delta}{2} + K_n \sqrt{K_n L_\delta} + \left(1 + 2\sqrt{\frac{K_n}{L_\delta}}\right) \ln \frac{8}{\delta}.$$

□